

QUANTUM-SECURED MAIL SYSTEM WITH QKD AND NO-CLONING PROPERTY

AI19711 PROJECT REPORT-II REPORT

Submitted by

MAHALAXMI VN [211501047]

LEKSHMI RAMAN KV [211501505]

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



**RAJALAKSHMI ENGINEERING COLLEGE
(AUTONOMOUS), CHENNAI-602 105
APRIL 2025**

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University Chennai)

BONAFIDE CERTIFICATE

Certified that this Phase-II Thesis titled “**QUANTUM SECURED MAIL SYSTEM WITH QKD AND NO-CLONING PROPERTY**” is The Bonafide work of **MAHALAXMI VN (211501047), LEKSHMI RAMAN KV (211501505)**, who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported here in does form part of any other thesis or dissertation on the basis of which a degree or award was confirmed on an earlier occasion on this or any other candidate.

Dr. K. SEKAR

Professor and Head

Department of Artificial Intelligence
and Machine Learning
Rajalakshmi Engineering College
Chennai- 602 105

Mrs. Akshaya V

Assistant Professor

Department of Artificial Intelligence
and Machine Learning
Rajalakshmi Engineering College
Chennai- 602 105

Submitted for the project viva-voce examination held on _____

Internal Examiner

External Examiner

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- To promote research activities amongst the students and the members of faculty that could benefit the society.
- To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES

PEO I: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO II: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO III: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO IV: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAM OUTCOMES (Pos)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate:

PSO 1: Foundation Skills: Ability to understand, analyze and develop

computer programs in the areas related to algorithms, system software, web design, AI and ML, DL, Data Science, and networking for efficient design of computer-based systems of varying complexity. Familiarity and practical competence with a broad range of programming language, tools and open-source platforms.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To develop the ability to solve a specific problem right from its identification and literature review till the successful solution of the same.
- To train the students in preparing project reports and to face reviews and viva voce examination.

COURSE OUTCOME

- On completion the students can able to execute the proposed plan and identify and overcome the bottle necks during each stage.
- On Completion of the project work students will be in a position to take

up any challenging practical problems and find solution by formulating proper methodology.

- Students will obtain a hands-on experience in converting a small novel idea / technique into a working model / prototype involving multi-disciplinary skills and / or knowledge and working in at team.
- Students will be able to interpret the outcome of their project.

CO-PO-PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1,2 or 3 are as defined below:

1: Slight (low) 2: Moderate (Medium) 3: Substantial (High)

No correlation: “-“

ABSTRACT

Quantum computing advancement poses risks to traditional cryptographic systems because it can break RSA and ECC encryption methods. The appearance of quantum threats creates a significant barrier to protect the confidentiality and integrity of information exchanged through email. The current email security protocols lack sufficient strength to defend against such attacks which demonstrates the immediate requirement for improved security protocols. The Quantum-Secured Mail System addresses this challenge by using Quantum Key Distribution (QKD) together with the No-Cloning Theorem to enhance email communication security. The quantum states used in QKD enable secure key distribution because any interception produces detectable anomalies based on the Heisenberg Uncertainty Principle. The No-Cloning Theorem provides additional security protection by making it impossible to duplicate quantum states which ensures data integrity. The system establishes a high standard for digital security through its implementation of proactive key exchange methods including BB84 and real-time intrusion detection for enhanced protection against unauthorized access in quantum technology.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman, **Mr. S. MEGANATHAN, B.E, F.I.E.**, our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, and our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. K. SEKAR, M.E., Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Machine Learning for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, Mrs. **AKSHAYA V, M.E.**, and Project Coordinator, **Dr. S. POONKUZHALI, M.E., Ph.D.**, Professor, Department of Artificial Intelligence and Machine Learning. Rajalakshmi Engineering College for their valuable guidance throughout the course of the project.

Lekshmi Raman KV- 211501505

Mahalaxmi VN - 211501047

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	viii
	ACKNOWLEDGEMENT	ix
	LIST OF FIGURES	xiv
	LIST OF TABLES	xiii
	LIST OF ABBREVIATIONS	xv
	LIST OF SYMBOLS	xvi
1	INTRODUCTION	1
	1.1 MOTIVATION	4
	1.2 SCOPE	6
	1.3 OBJECTIVES	9
2	LITERATURE SURVEY	10
	2.1 LITERATURE REVIEW	10
	2.1.1 FOUNDATIONS AND EARLY ADVANCEMENTS IN QUANTUM CRYPTOGRAPHY	10
	2.1.2 EMERGING TRENDS IN PROTOCOL DESIGN AND QUANTUM PRINCIPLES	11
	2.1.3 ADVANCES IN QUANTUM KEY DISTRIBUTION (QKD) TECHNOLOGIES	11
	2.1.4 APPLICATION-FOCUSED STUDIES IN SECURE COMMUNICATION	12

	2.1.5 CHALLENGES AND LIMITATIONS	12
	IN QUANTUM SECURED SYSTEM	
	2.1.6 ENHANCEMENTS IN PROTOCOLS	13
	AND SYSTEM ARCHITECTURE	
3	SYSTEM DESIGN	14
	3.1 DEVELOPMENT ENVIRONMENT	14
	3.1.1 SYSTEM ARCHITECTURE	14
	3.1.2 USE CASE DIAGRAM	17
	3.2 SYSTEM REQUIREMENTS	18
	3.2.1 HARDWARE SPECIFICATIONS	18
	3.3.2 SOFTWARE SPECIFICATIONS	19
4	METHODOLOGY	21
	4.1 GENERAL	21
	4.2 METHODOLOGY USED	22
	4.3 MODULE DESCRIPTIONS	30
	4.3.1 QMAIL SENDER MODULE	30
	4.3.2 QUANTUM KEY GENERATION AND	31
	DISTRIBUTION MODULE	
	4.3.3 QKD ENCRYPTION MODULE	31
	4.3.4 KEY VERIFICATION AND	32
	AUTHENTICATION	
	4.3.5 SECURE MAIL TRANSMISSION AND	32
	RECEIVER MODULE	
	4.3.6 DECRYPTION AND MAIL VIEWING	32
5	RESULTS AND DISCUSSIONS	34
	5.1 RESULTS	34

	5.2 DISCUSSIONS	44
6	CONCLUSION AND FUTURE ENHANCEMENTS	47
	6.1 CONCLUSION	47
	6.2 FUTURE ENHANCEMENTS	48
	APPENDIX	49
	PAPER PUBLICATION	49
	REFERENCES	55

LIST OF TABLES

FIGURE NO.	TITLE	PAGE NO.
3.2.1.1	HARDWARE SPECIFICATION	19
3.2.2.1	SOFTWARE SPECIFICATION	20
5.2.1	PERFORMANCE ANALYSIS	44

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1.1	SYSTEM ARCHITECTURE OF QMAIL	14
3.1.2	USE CASE MODEL	17
5.1.1	THE GENERATION AND ENCRYPTION OF A KEY	34
5.1.2	THE SHARED KEY	34
5.1.3	THE DECRYPTED KEY AND THE MESSAGE	36
5.1.4	QKD METRICS AND THE PROCESSING TIME	36
5.1.5	CLONING ATTEMPT AND THE REGENERATION OF A NEW KEY	39
5.1.6	QBER CHART	40
5.1.7	QMAIL UI	41
5.1.8	LOGIN PAGE	42
5.1.9	HOME PAGE	42
5.1.10	COMPOSED A MAIL	43
5.1.11	QMAIL INBOX	43
5.1.12	DECRYPTED MESSAGE	44

LIST OF ABBREVIATIONS

QKD	Quantum Key Distribution
AES	Advanced Encryption Standard
QSE	Quantum Secured Email
HTTPS	Hypertext Transfer Protocol
UI	User Interface
SQL	Structured Query Language
BB84	Bennett-Brassard Protocol
EDM	Eavesdrop Detection Mechanism
KMG	Key Management Generation
QC	Quantum Channel
CI	Cipher Text

LIST OF SYMBOLS

Δx – Uncertainty in position.

Δp – Uncertainty in momentum.

$\hbar$ (h-bar) – Reduced Planck's constant, where $\hbar = h/2\pi$.

$\geq$ – Greater than or equal to.

$|\psi\rangle$ – Quantum state (wave function).

α (alpha) – Probability amplitude associated with the state $|0\rangle$.

β (beta) – Probability amplitude associated with the state $|1\rangle$.

$|0\rangle$ and $|1\rangle$ – Basis states representing quantum superposition.

$|\alpha|^2$ – Probability of the system being in state $|0\rangle$.

$|\beta|^2$ – Probability of the system being in state $|1\rangle$.

$=$ – Equality sign, indicating that the sum of probabilities is equal to 1.

$\otimes$ – Tensor product of Quantum states.

$|e\rangle$ – Quantum state

CHAPTER 1

INTRODUCTION

Email has greatly changed the way that people communicate, becoming a vital tool in personal and professional fields. Its effectiveness, convenience, and quickness have established it as the most popular means of communication, used for everything from casual chats to important business exchanges. Over time, email has become a crucial part of the modern communication system, serving millions of users worldwide. Still, the ease of use and popularity of email have also made it a top choice for cyber attackers. Email systems are at risk of various security threats, such as phishing, man-in-the-middle attacks, and unauthorized access due to their open nature. These weaknesses have resulted in major leaks of confidential data, underscoring the critical importance of strong security measures.

The Evolution of Email Security:

In response to the increasing quantity of cyber threats, different security strategies have been created. Encryption has become the foundation of email security, with many considering it a key component. Encryption involves transforming clear text into unintelligible ciphertext to guarantee that solely authorized individuals can reach the material. This is done using cryptographic keys—specific information that enables encryption and decryption of data. Encryption in conventional email systems usually depends on algorithms like RSA (Rivest–Shamir–Adleman) and ECC (Elliptic Curve Cryptography). The RSA algorithm, one of the first public-key encryption methods, has been extensively employed to secure emails. To combat rising cyber threats, especially those targeting digital communications, email security has become a major focus. One of the most widely adopted solutions is encryption, which ensures that sensitive data remains confidential during transmission. It protects against unauthorized access by

converting readable messages into scrambled code. Among various cryptographic techniques, public-key encryption stands out as a reliable method for secure communication. This process requires creating two keys: one public key for encrypting data and one private key for decrypting data. The security of the RSA encryption method is ensured by the challenge of factorizing large numbers, which is currently difficult to achieve with available computational power. However, Elliptic Curve Cryptography (ECC) provides comparable security to RSA, yet with much smaller key lengths, leading to quicker calculations and decreased processing requirements.

Digital Signatures and Authentication:

Besides encryption, email systems make use of digital signatures to confirm the sender's identity and maintain the message's integrity. A digital signature is a method of verifying the legitimacy of digital messages or documents using mathematical algorithms. During email exchanges, the sender employs a personal key to sign a message, and the receiver utilizes the sender's public key to authenticate the signature. This method aids in preventing impersonation attacks, in which a malicious actor may try to send fake emails pretending to be a trustworthy sender. By using digital signatures, email systems can ensure the message has not been changed and comes from the stated sender.

Challenges with Traditional Encryption:

Even though RSA, ECC, DES and digital signatures are effective, conventional encryption techniques encounter major obstacles. One of the biggest concerns is the fast evolution of technology, specifically the emergence of quantum computing. In contrast to classical computers that use binary bits (0s and 1s), quantum computers utilize quantum bits, or qubits, that are able to be in multiple states at the same time because of superposition

principles. This capacity allows quantum computers to complete intricate calculations much faster than classical systems. The advancement of quantum computing presents a significant challenge to conventional encryption techniques. A quantum computer capable of processing power could potentially crack RSA and ECC encryption by quickly factorizing large numbers, a task that classical computers cannot currently achieve. Concerns regarding the security of traditional encryption have increased due to the potential impact of Shor's algorithm, a quantum algorithm capable of factorizing large integers quickly. If Shor's algorithm is effectively applied by a quantum computer, it would make RSA encryption outdated, putting at risk all communications secured by this conventional means.

Emergence of Quantum-Resistant Cryptography:

To deal with the dangers brought by quantum computing, there is an increasing focus on quantum-resistant cryptographic techniques. These encryption algorithms are newly created to resist quantum computer attacks. Quantum-resistant cryptography, sometimes referred to as post-quantum cryptography. Cryptography depends on challenging mathematical problems that are difficult to solve, even for a quantum computer. Potential solutions being investigated include lattice-based cryptography, hash-based cryptography, and multivariate polynomial cryptography. These techniques aim to offer a high level of security that can withstand traditional and quantum computing risks.

Quantum Key Distribution (QKD) and Its Promise:

Alongside quantum-resistant cryptography, Quantum Key Distribution (QKD) has also emerged as an innovative method for securing communications during the quantum era. Contrary to traditional encryption,

QKD utilizes the laws of quantum mechanics instead of computational complexity to protect cryptographic key exchanges. In Quantum Key Distribution (QKD), data is sent using quantum states, like photon polarization, which are highly susceptible to being observed. As per the principles of quantum mechanics, any effort to intercept or measure these quantum states would change them, causing eavesdropping to be noticeable. This feature ensures that QKD is always secure, as any unauthorized interception can be quickly detected and resolved.

The No-Cloning Theorem and Its Role in QKD:

Another important concept in quantum cryptography is the no-cloning theorem, which asserts that duplicating an unidentified quantum state perfectly is not feasible. This theorem is crucial for the security of QKD as it stops hackers from intercepting and replicating quantum keys unnoticed. The no-cloning theorem provides additional protection for quantum key exchange, ensuring the communication's integrity is maintained. If someone tries to steal the quantum key, noticeable abnormalities will be created in the quantum states, letting the rightful parties respond accordingly.

1.1 MOTIVATION

1.1.1 Motivation for Developing a Quantum-Secured Mail System

With the rapid growth of digital communication, traditional email systems face an increasing number of cyber threats, including hacking, phishing, and man-in-the-middle attacks. Conventional encryption methods such as RSA and SSL/TLS are vulnerable to quantum computing advancements, which can break these encryption systems. As cybersecurity threats become more sophisticated, there is a critical need to enhance the security of communication systems, especially email systems that handle sensitive data.

Classical cryptographic systems rely on mathematical complexity for security, but these methods are becoming less reliable with the emergence of quantum computers. Quantum computers can easily break large integer factorization and discrete logarithm-based encryption methods, which form the backbone of traditional encryption. This potential vulnerability raises the need for exploring new cryptographic models that can withstand quantum attacks and ensure long-term security. Quantum Key Distribution (QKD) offers a revolutionary solution by using the principles of quantum mechanics to generate and distribute cryptographic keys securely. Protocols such as BB84 ensure that any attempt to intercept the key introduces detectable errors, alerting legitimate parties (Alice and Bob) about potential threats. Additionally, the No-Cloning Theorem ensures that an adversary (Eve) cannot copy the quantum state without introducing disturbances, making eavesdropping impossible. Email remains one of the most widely used communication mediums for both personal and professional interactions. However, its security is often compromised due to weak encryption protocols. Implementing a Quantum-Secured Email System that integrates QKD can significantly improve the security of sensitive email communications. By leveraging QKD to generate and distribute encryption keys, the system ensures that emails remain secure and unaltered, even in the presence of an adversary. Developing a secure email system with QKD is not just a theoretical advancement but has immense real-world implications. Industries such as finance, healthcare, government, and defense handle confidential information that requires the highest level of security. By incorporating QKD into email communication, these industries can ensure data privacy and integrity, minimizing the risks associated with quantum threats. The motivation to undertake this project stems from the realization that current email security systems may not be future-proof against emerging threats. By exploring the application of QKD and the No-Cloning Theorem,

this project aims to bridge the gap between theoretical quantum cryptography and practical email security. The ultimate goal is to create a system where only Alice and Bob can securely exchange messages, eliminating any possibility of unauthorized interception.

1.2 SCOPE

The Quantum-Secured Email System using Quantum Key Distribution (QKD) and the No-Cloning Theorem aims to enhance the security of email communication by implementing advanced quantum cryptographic techniques. This project focuses on developing a secure email system where encryption and decryption are safeguarded using QKD, ensuring that only authorized users can access the transmitted information. The system prevents any possibility of eavesdropping or interception by leveraging the No-Cloning Theorem, which makes it impossible to clone quantum states without introducing detectable errors.

The scope of the project includes the following major components:

1.2.1 Quantum Key Distribution (QKD) Implementation:

The project involves designing and implementing a secure key exchange mechanism using the BB84 protocol, a foundational method in Quantum Key Distribution (QKD). This protocol ensures that any interception attempts by an adversary introduce detectable errors in the quantum state, enabling the legitimate parties, Alice and Bob, to identify unauthorized access. Once a secure key is successfully established, it is used as a symmetric encryption key to protect the confidentiality and integrity of email communication between users.

1.2.2 Secure Email Transmission:

The system is developed to perform encryption and decryption of email

content using AES encryption, with the encryption key generated through Quantum Key Distribution. This ensures that the email content remains confidential and secure during transmission. The encrypted emails are transmitted securely to the intended recipient, who uses the same quantum-generated key to decrypt and read the message. Additionally, integrity checks are implemented to detect any tampering or unauthorized modifications during transmission, ensuring the authenticity and reliability of the communication.

1.2.3 Website Development for Email System:

A secure and user-friendly web-based email platform was created to enable the sending and receiving of encrypted emails. Key modules were developed to enhance the platform's functionality, including a secure user authentication system for login and registration, a "Compose Email" interface for drafting and encrypting messages, and dedicated sections for viewing received (Inbox) and sent (Sent Items) emails. Additionally, a Key Exchange Dashboard was implemented to display the status of key generation and exchange. The frontend was developed using HTML, CSS, and JavaScript, while the backend was built with Flask (Python) to ensure smooth and efficient functionality.

1.2.4 Backend and Database Integration:

The backend system was implemented to manage critical functions such as email encryption, key management, and secure message transmission. MySQL/SQL databases were used to securely store user information, encrypted emails, and key data. To ensure the protection of sensitive data during communication, a secure channel was established between the frontend and backend, maintaining confidentiality and integrity throughout the data exchange process.

1.2.5 Attack Simulation and Detection:

A Man-in-the-Middle (MITM) attack was simulated to demonstrate the effectiveness of Quantum Key Distribution (QKD) in detecting unauthorized interception attempts. The simulation involved analyzing the error rates introduced by an eavesdropper (Eve) during the key exchange process. The results confirmed that QKD could successfully identify these anomalies, as the presence of Eve introduced detectable disturbances in the quantum states. Consequently, any intercepted or compromised keys were discarded, ensuring the security and integrity of the communication.

1.2.6 Visualization and Performance Analysis:

Visual representations were generated to analyze the overall performance of the system. These visualizations included key generation timelines, error rates observed during the key exchange process, and comparisons of encryption and decryption times. Additionally, graphs were used to illustrate the detection of eavesdropping attempts, showcasing the system's ability to identify unauthorized interceptions. The performance and security of the quantum-secured email system were also compared with traditional email encryption protocols, highlighting the enhanced protection and reliability provided by Quantum Key Distribution (QKD).

1.2.7 Comparison with Traditional Email Systems:

The project highlights the limitations of traditional encryption methods like RSA and DES in securing email communication, particularly their vulnerability to future quantum attacks and key compromise. In contrast, the integration of Quantum Key Distribution (QKD) and the No-Cloning Theorem provides a significantly higher level of security. QKD ensures that any attempt to intercept the key introduces detectable anomalies, while the No-Cloning Theorem guarantees that quantum states and therefore the keys

cannot be copied or reused by an adversary. This combination offers robust protection against both current and future threats, ensuring the confidentiality and integrity of email communication.

1.3 OBJECTIVES

- To detect eavesdropping activities during the key exchange process using the No-Cloning Theorem.
- To enhance user privacy and data confidentiality without compromising the usability of the email system.
- To integrate Quantum Key Distribution (QKD) into email communication for providing highly secure key exchange.
- To ensure seamless compatibility of the quantum-secured email system with popular email services like Gmail and Yahoo Mail.

CHAPTER 2

LITERATURE SURVEY

We have conducted a thorough literature survey to gain insights into advancements in quantum communication and cryptographic techniques, with a focus on enhancing security through Quantum Key Distribution (QKD). Our survey highlights improvements in key exchange protocols, error correction mechanisms, and secure communication models. It also explores methods to mitigate challenges such as photon loss, device imperfections, and scalability issues. Additionally, we analyzed the integration of quantum technologies with existing communication infrastructures to strengthen data security. These insights provide a solid foundation for the secure and efficient implementation of our Quantum-Secured Mail System.

2.1 LITERATURE REVIEW

2.1.1 Foundations and Early Advancements in Quantum Cryptography

Jorstad et al., *Cryptographic Algorithm Metrics* (1997), presented early methods to evaluate cryptographic algorithms, laying essential groundwork for later developments in quantum-secured systems. Zeng et al., *Quantum Communication* (2000), focused on advancements in QKD and quantum state manipulation, establishing key directions for secure communication. Kanamori et al., *Authentication Protocol Using Quantum Superposition States* (2009), introduced quantum superposition in authentication protocols, contributing significantly to early-stage implementation ideas. Gisin et al., *Quantum Computing* (2007), examined integration of QKD into classical

communication networks, marking a transition toward hybrid security systems. Qiao et al., *Simulation of BB84 Quantum Key Distribution in Depolarizing Channel* (2011), provided an analytical model for BB84 protocol behavior under noisy channels, emphasizing practical limitations in early quantum channels.

2.1.2 Emerging Trends in Protocol Design and Quantum Principles

Tamaki et al., *One-Step Quantum Secure Direct Communication* (2014), developed security proofs for one-step QKD protocols, streamlining communication efficiency while preserving theoretical integrity. Mailloux et al., *Performance Evaluations of QKD System Architecture* (2015), evaluated different QKD designs, offering insight into architectural improvements for scalable quantum systems. Diamanti et al., *Practical Challenges in Quantum Key Distribution* (2016), outlined key technological and environmental constraints limiting widespread adoption of QKD. Ortigoso et al., *Twelve Years Before the Quantum No-Cloning Theorem* (2018), reflected on the pre-theoretical ideas that inspired the no-cloning theorem, framing it within the historical progression of quantum logic. Cacciapuoti et al., *Quantum Internet Networking Challenges* (2019), examined challenges in deploying the quantum internet, particularly in distributed quantum computing.

2.1.3 Advances in Quantum Key Distribution (QKD) Technologies

Pirandola et al., *Advances in Quantum Cryptography* (2020), offered a comprehensive survey of QKD implementation in both fiber-optic and free-space systems, marking a leap toward real-world readiness. Bluhm et al., *Position-Based Cryptography: Single-Qubit Protocol Secure Against Multi-Qubit Attacks* (2021), introduced a lightweight protocol using position verification, withstanding advanced multi-qubit attack models. Portmann et al., *Security in Quantum Cryptography* (2022), reviewed modern quantum

cryptographic strategies, unifying theoretical and practical views on system robustness and authentication. Almashaqbeh et al., *Unclonable Cryptography: A Tale of Two No-Cloning Paradigms* (2023), explored novel applications of the no-cloning theorem in protocol development, drawing parallels between quantum hardware limitations and security enhancements. Dejpasand et al., *Research Trends in Quantum Computers by Focusing on Qubits as Their Building Blocks* (2023), focused on hardware evolution and the foundational role of qubits in supporting next-gen quantum security protocols.

2.1.4 Application-Focused Studies in Secure Communication

Mailloux et al., *Performance Evaluations of QKD System Architecture* (2015), presented one of the first empirical studies evaluating complete QKD system designs, aiding stakeholders in selecting practical configurations. Cacciapuoti et al., *Quantum Internet Networking Challenges* (2019), applied theoretical models to distributed quantum environments, proposing frameworks for long-distance secure communication. Pirandola et al., *Advances in Quantum Cryptography* (2020), also addressed practical deployments in fiber and satellite links, pushing theoretical innovations closer to implementation. Vasani et al., *Embracing the Quantum Frontier* (2024), summarized recent advances in communication protocols, including entanglement-based exchanges and ultra-secure quantum email proposals.

2.1.5 Challenges and Limitations in Quantum-Secured Systems

Diamanti et al., *Practical Challenges in Quantum Key Distribution* (2016), highlighted system vulnerabilities, like photon loss and channel noise, that reduce reliability and distance. Almashaqbeh et al., *Unclonable Cryptography* (2023), underlined challenges in physical realization of unclonable states, noting that implementation complexity may hinder

scalability. Dejpasand et al., *Research Trends in Quantum Computers* (2023), also acknowledged the technological gap in creating stable and error-resistant qubit systems. Vasani et al., *Embracing the Quantum Frontier* (2024), emphasized unresolved challenges in harmonizing quantum systems with classical infrastructure, especially when applied to communication networks such as quantum email platforms.

2.1.6 Enhancements in Protocols and System Architectures

Tamaki et al., *One-Step Quantum Secure Direct Communication* (2014), proposed a simplified QKD model with direct communication security, backed by formal proofs. Mailloux et al., *Performance Evaluations of QKD System Architecture* (2015), evaluated real-time QKD systems, measuring performance under practical constraints and helping bridge theoretical design with implementation. Diamanti et al., *Practical Challenges in Quantum Key Distribution* (2016), reviewed obstacles in real-world QKD, such as photon detection inefficiency, system calibration, and distance limitations. Ortigoso et al., *Twelve Years Before the Quantum No-Cloning Theorem* (2018), revisited foundational ideas that predated the no-cloning theorem, providing context to one of quantum cryptography's key principles.

CHAPTER 3

SYSTEM DESIGN

3.1 DEVELOPMENT ENVIRONMENT

3.1.1 SYSTEM ARCHITECTURE

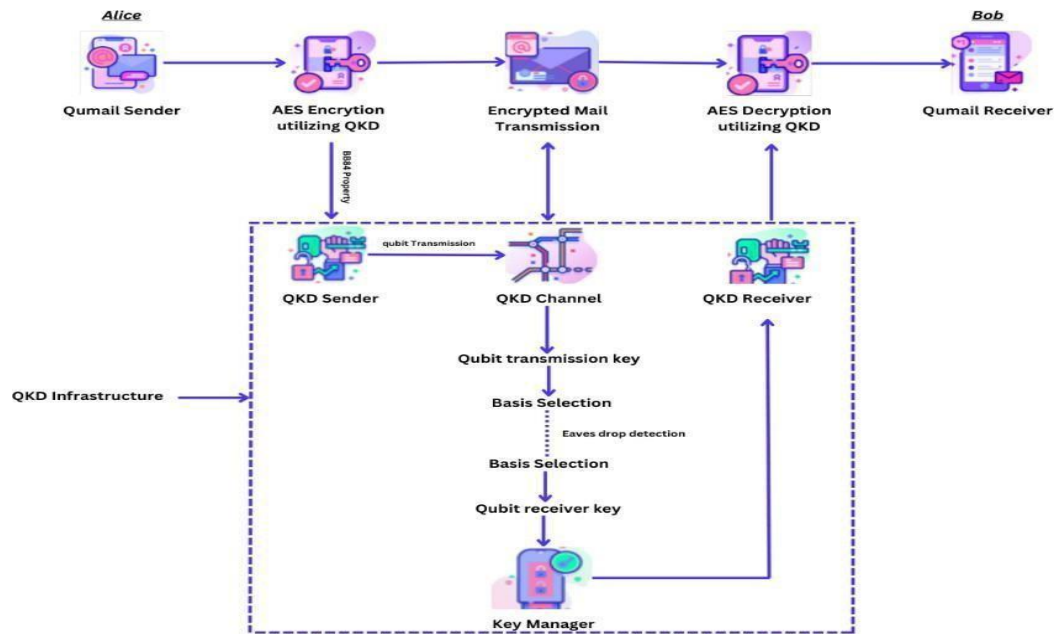


Fig. 3.1.1 System Architecture of Qmail

In the quantum-secured email system concept, Alice (the sender) starts an email using the "Qmail Sender" module and then encrypts it with AES encryption. The encryption key is produced through Quantum Key Distribution (QKD) with the BB84 protocol to guarantee its uniqueness and security. After receiving the encrypted email, Bob uses the "Qmail Receiver" module and the corresponding QKD-generated key to decrypt it, ensuring that only he can read the content. The QKD infrastructure plays a crucial role in the system, ensuring a secure communication channel for exchanging keys and using the BB84 protocol to prevent spying. Alice's QKD Sender creates and sends qubits over the QKD channel, and Bob's QKD Receiver measures

the qubits to recreate the encryption key. While sending information, Alice and Bob choose random measurement bases to identify any interference or eavesdropping. After generating a secure key, the Key Manager will securely maintain and oversee it for future encryption and decryption processes, guaranteeing the privacy and authenticity of email exchanges.

The proposed system consists of the following core modules:

The Qmail Sender and Receiver Modules are the endpoints of the quantum-secured communication system. The Qmail Sender, operated by the sender (Alice), is responsible for composing the email and initiating the encryption process. On the other side, the Qmail Receiver, used by the recipient (Bob), decrypts and reads the email after secure key exchange and encryption. These modules serve as the user interfaces of the system and are built to seamlessly integrate classical email operations with quantum encryption mechanisms. The AES Encryption and Decryption Modules utilizing QKD form the core of the system's data protection. AES (Advanced Encryption Standard) is used for encrypting and decrypting the actual message content due to its efficiency and reliability. However, instead of using a traditional static key, this system uses keys generated by the QKD (Quantum Key Distribution) process. This ensures that the encryption keys are random, secret, and not susceptible to interception, thanks to the principles of quantum mechanics. The QKD Infrastructure Module includes several key components: the QKD Sender, QKD Receiver, QKD Channel, and Key Manager. The QKD Sender transmits quantum bits (qubits) using a quantum channel. These qubits encode random bits of information that will form the secret encryption key. The QKD Receiver measures these qubits using selected bases. Any attempt to intercept the qubits during transmission alters their state, enabling eavesdropping detection. Once verified, the QKD Receiver sends the

matched key bits to the Key Manager for secure storage and synchronization between Alice and Bob.

Finally, the Key Manager Module plays a central role in managing the cryptographic keys derived from the QKD process. It performs operations like matching the qubit transmission and receiver keys, handling basis selection, filtering out corrupted or mismatched bits, and generating the final secure key. This key is then passed to the AES encryption and decryption modules at both ends. This integration of quantum and classical cryptographic techniques ensures end-to-end confidentiality, authenticity, and future-proof security for email communication.

3.1.2 USE CASE DIAGRAM

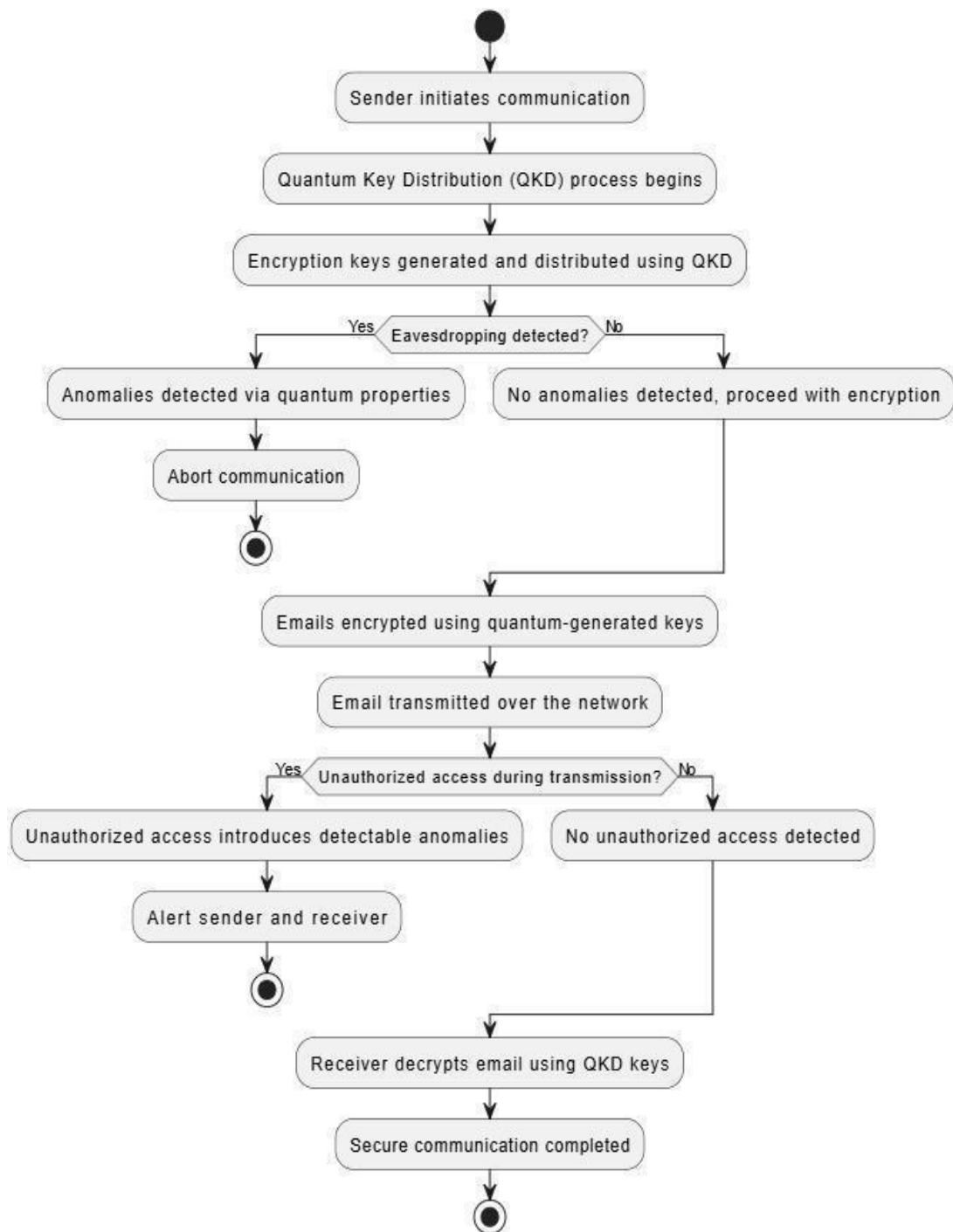


Fig. 3.1.2 Use Case Model of Qmail

Fig 3.1.2 This illustration shows how a quantum-protected email system operates with Quantum Key Distribution (QKD) in place. The sender

initiates the process by creating a quantum key with QKD. The quantum key is sent through a quantum channel to the recipient, guaranteeing the creation of a secure key. The next stage includes ensuring the key's integrity by verifying if it corresponds to the sender's key, done by the recipient. Assuming the key is legitimate, the sender will go on to encrypt the email with the quantum-protected key before transmitting it to the recipient through a conventional channel. Throughout the Quantum Key Distribution phase, constant surveillance is conducted to identify any suspicious activities that may indicate eavesdropping. If no unusual activity is found, the encrypted email is delivered to the recipient, who is able to decode it using the identical quantum key, thus maintaining the security of the email. In case of any irregularity in key transfer, the procedure is halted to avoid unauthorized entry. At last, the recipient accesses the decrypted email, finalizing the secure communication process. The illustration distinctly delineates the quantum and classical communication pathways, emphasizing the secure sending and validation of keys.

3.2 SYSTEM REQUIREMENTS

3.2.1 HARDWARE SPECIFICATIONS

The section outlines the recommended hardware specifications necessary for the efficient and secure deployment of the Quantum-Secured Email System. The system demands a stable computing infrastructure to perform quantum key generation, AES encryption, and secure mail delivery without performance bottlenecks. The listed components are selected to ensure smooth operation of cryptographic algorithms and to support quantum simulation libraries like High-speed processing, adequate memory, and reliable storage are essential to maintain the performance and responsiveness of the system.

Table .3.2.1.1 Hardware specification table

COMPONENT	SPECIFICATIONS
Processor	Intel Core i7 or above
RAM	16GB or above (DDR4)
Storage	256GB SSD or higher
Processor Frequency	2.6 GHz or above

3.2.2 SOFTWARE SPECIFICATIONS

The section outlines the necessary software components required to develop and implement the Quantum-Secured Email System. The system is designed to work in a Windows 11 environment and utilizes a combination of web development tools, quantum computing libraries, and secure communication protocols. The selection of tools ensures seamless integration between quantum key distribution (QKD), AES encryption, and secure mail transmission. The server-side functionalities are built using Flask-Mail, and the backend encryption processes leverage libraries like Cryptography. MySQL, accessed via XAMPP, handles database management effectively.

Table .3.2.2.1 Software Specification Table

COMPONENT	SPECIFICATIONS
Operating System	Windows 11
Development Tools	VS Code, Anaconda, Jupyter Notebook, XAMPP, SQL
Email Server Software	Flask-Mail (SMTP)
Encryption Libraries	Flask, Flask-CORS, PyMySQL, Flask-Mail, Cryptography, Hashlib
Database Management	MySQL (XAMPP),SQL

CHAPTER 4

METHODOLOGY

4.1 GENERAL

The Quantum-Secured Email System is designed to create a secure and reliable communication platform by blending classical and quantum encryption methods. The system consists of multiple important parts that collaborate to guarantee the privacy and security of email exchanges. The core of the design features the Quantum Key Distribution (QKD) module, enabling the safe creation and sharing of cryptographic keys. The BB84 protocol is utilized by the system to send qubits through a quantum channel, making it possible to detect any interception of the keys because of the quantum states' sensitivity. This configuration allows for a secure exchange of keys between the sender and recipient, serving as the foundation for encrypting email messages.

The layout comprises modules such as the Qmail Sender and Qmail Receiver, which handle creating, transmitting, and receiving emails. These modules utilize AES encryption for encrypting and decrypting data, with keys coming from the QKD system. The system includes a strong Key Management system to securely store and control encryption keys, guaranteeing they can only be accessed by authorized users. The system's focus is on detecting eavesdropping in real-time while conducting Quantum Key Distribution, by getting rid of compromised keys and starting a new exchange if any interference is found. Furthermore, compatibility with traditional email protocols (SMTP, IMAP) guarantees the system can function smoothly with current infrastructure, reducing interruptions and improving security. The system also includes eavesdropping detection mechanisms that work in real-time while the key exchange process is happening. This includes

observing the quantum channel for any disruptions that could signal an interception effort. If a flaw is found, the compromised key will be eliminated, and a new secure key exchange will be started to prevent email content from being encrypted with a potentially compromised key. Moreover, the Email Encryption and Decryption Modules are essential components in the system's design. The modules encrypt email content with AES using quantum-generated keys, ensuring only the intended recipient can decrypt and read the message. In order to ensure strong security measures, the system combines Classical Encryption for efficient data handling and Quantum Key Distribution for secure key management in a Hybrid Encryption Approach.

4.2 METHODOLOGY USED

Implementing the Quantum-Secured Email System requires a systematic approach utilizing quantum principles and advanced encryption methods to guarantee secure communication through email. The central method focuses on Quantum Key Distribution (QKD), which produces a confidential key utilizing quantum characteristics such as photon polarization. This key is essential for securely encrypting and decrypting email content. The Quantum Key Distribution (QKD) detects any potential eavesdropping during key exchange, protecting the communication's confidentiality. Following the QKD phase, the shared key is utilized in the Email Encryption Module, which uses the Advanced Encryption Standard (AES) algorithm to secure the email content. The encrypted email is sent securely using standard internet protocols like SMTP. The recipient then uses the same shared key to decrypt the message through the decryption module. A user-friendly web application allows easy sending and receiving of encrypted emails. All user data and keys are stored securely with strong encryption.

The secure encrypted email is then safely sent using standard internet protocols, maintaining data integrity while being transmitted. The recipient employs the identical shared key for the Email Decryption Module to reverse the encryption procedure and access the initial email text. The Web Application is designed to be a user interface that allows for encrypted emails to be sent and received in a simple and easy-to-use manner. The user data, keys, and metadata are stored securely in the backend database, with encryption measures in place to safeguard sensitive information. The Agile Methodology is utilized for the development and integration of these modules, enabling iterative testing and feedback to enhance the system's functionality and security. Every stage is carefully validated to guarantee the system's reliability and strength against possible quantum and classical risks.

4.2.1 Quantum Key Distribution (QKD) Implementation

QKD ensures secure key exchange between sender (Alice) and receiver (Bob) by leveraging the principles of quantum mechanics. The BB84 protocol is used to generate and distribute encryption keys. Alice sends randomly polarized photons to Bob using four polarization states: horizontal, vertical, diagonal, and anti-diagonal. Bob measures the incoming photons with randomly chosen bases. Alice and Bob compare a subset of their measurement results over a public channel to verify consistency. Errors due to eavesdropping (Eve) are detected using the No-Cloning Theorem, which prevents perfect duplication of quantum states. Error correction is done using ECC (Error Correction Codes) and privacy amplification to minimize Eve's information. The Heisenberg Uncertainty Principle ensures that any attempt to intercept the quantum channel introduces errors in the transmission. This principle states that certain pairs of quantum properties, such as position and momentum, cannot be precisely measured simultaneously, which is expressed mathematically as:

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (1)$$

Where:

Δx - (Uncertainty in Position)

Δp - (Uncertainty in Momentum)

$\hbar$ (h-bar) - (Reduced Planck's Constant)

Here, the equation (1) states that:

The Δx represents the uncertainty in position, and Δp denotes the uncertainty in momentum. The term $\hbar$ is the reduced Planck's constant, defined as $\hbar = h/2\pi$, where h is Planck's constant (6.626×10^{-34} J s). This principle highlights that the more precisely the position of a particle (like an electron) is known, the less precisely its momentum can be determined, and vice versa. The product of these uncertainties can never be smaller than $\hbar/2$, establishing a fundamental limit to measurement accuracy at the quantum level. This concept is a cornerstone of quantum mechanics and explains the inherent randomness in the behavior of microscopic particles. Additionally, the principle of quantum superposition is applied during the transmission of qubits. A quantum state can exist in multiple states until a measurement collapses it to one definite state. Mathematically, the quantum superposition state can be represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (2)$$

Where:

$|\psi\rangle$ - (Quantum State)

$|0\rangle$ and $|1\rangle$ - (Basis States)

α and β - (Probability Amplitudes)

where α and β are probability amplitudes that satisfy:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (3)$$

Application in BB84 Protocol (QKD):

In the BB84 protocol (the most commonly used QKD protocol), Alice prepares and sends qubits in different bases.

If Alice sends the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad (4)$$

The probabilities are:

$$P(0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad \text{and} \quad P(1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (5)$$

Normalization check:

$$\frac{1}{2} + \frac{1}{2} = 1 \quad (6)$$

Error Detection in QKD:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (7)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{with} \quad |\alpha|^2 + |\beta|^2 = 1 \quad (8)$$

This ensures that the transmitted qubits remain in a valid quantum state throughout the QKD process, allowing Alice and Bob to detect errors and ensure secure key generation.

4.2.2 Encryption and Decryption Using QKD Keys

After generating a secure key through QKD, the key is used for symmetric encryption and decryption of emails. The plaintext email is encrypted using the Advanced Encryption Standard (AES), a symmetric encryption algorithm known for its robustness. The encryption process involves several rounds of transformations, including substitution, permutation, and mixing, making it difficult for adversaries to break the encryption. The generated QKD key ensures that only Alice and Bob possess the correct key. Decryption occurs when Bob receives the encrypted message and uses the same symmetric key obtained through QKD. Without the correct key, even if an attacker intercepts the encrypted message, decryption is impossible. Encryption and decryption durations were recorded during simulations, with encryption time noted as 0.000002 seconds and decryption time at 0.000002 seconds, ensuring computational efficiency.

4.2.3 Detection of Eavesdropping and Error Correction

Eavesdropping detection is a critical component of QKD. Any attempt to intercept the quantum channel introduces detectable errors in the transmission. The No-Cloning Theorem guarantees that an adversary (Eve) cannot duplicate quantum states without introducing measurable disturbances. Error correction is performed using Error Correction Codes (ECC) to fix transmission errors caused by photon loss or noise. Error estimation is conducted, and if the error rate exceeds a predefined threshold, the communication is aborted to prevent key compromise.

The Error Rate is measured using the formula:

$$\text{Error Rate (ER)} = (N_{\text{error}} / N_{\text{total}}) \times 100$$

where:

- N_{error} is the number of erroneous qubits received.
- N_{total} is the total number of transmitted qubits.

$N_{\text{total}} = 12,000$ bits transmitted

$N_{\text{error}} = 2,000$ error bits

$$ER = (2,000 / 12,000) \times 100$$

$$(1/6) \times 100 \approx 16.67\%$$

In our experimental setup, the recorded error rate was 16.67% when eavesdropping was detected, prompting the system to regenerate a secure key.

4.2.4 Key Verification and Authentication

Key verification ensures that the keys generated through QKD are consistent between Alice and Bob. This process is conducted through a public authenticated channel, where any discrepancies between The keys are corrected using post-processing techniques such as reconciliation and error correction. Authentication is achieved using digital signatures to validate the source and ensure message integrity. The digital signature verifies that the key has not been tampered with, providing an additional layer of protection.

4.2.5 Secure Email Transmission Process

The secure email transmission process involves multiple layers of security to maintain confidentiality and authenticity. The steps include:

- Step 1: Alice composes and encrypts the email using the symmetric key obtained via QKD.
- Step 2: The encrypted email is transmitted over a classical communication channel.
- Step 3: Bob receives the encrypted email and decrypts it using the

symmetric key.

- Step 4: Key authentication and verification ensure that no third party has tampered with the communication.

4.2.6 Error Handling and Protocol Security

Error handling is critical to maintaining the integrity of the encrypted communication. High error rates can compromise key integrity, so advanced error detection and correction protocols are implemented. Privacy amplification further reduces the chances of partial key leakage to adversaries. The BB84 protocol inherently detects eavesdropping by introducing errors when an adversary attempts to intercept communication. Secure authentication ensures that only Alice and Bob share the symmetric key. Additionally, Quantum Bit Error Rate (QBER) values are monitored to ensure that the error rate remains within acceptable limits.

The QBER is defined as:

$$\text{QBER} = (\text{Nerror} / \text{Ntotal}) \times 100$$

Nerror – Number of error bits

Ntotal – Total Number of bits transmitted

Transmitted Bits: Ntotal = 10,000

Error Bits Detected: Nerror = 164

$$\begin{aligned} \text{QBER} &= (164 / 10,000) \times 100 \\ &= 1.64\% \end{aligned}$$

In our system, the QBER was recorded at 1.64%, indicating a low error rate and high reliability in quantum key distribution.

4.2.7 Evaluation and Performance Metrics

The system's performance is evaluated based on four key metrics:

- Key Generation Rate: The number of secure keys generated per second.
 - Error Rate: The percentage of errors detected during key verification.
 - Transmission Distance: The maximum distance over which QKD remains secure.
 - Throughput and Latency: Efficiency of secure communication.
- The Total Processing Time (TPT), which is the sum of encryption and decryption times, is calculated as:

$$ET = \text{End Time of Encryption} - \text{Start Time of Encryption}$$

$$DT = \text{End Time of Decryption} - \text{Start Time of Decryption}$$

$$TPT = ET + DT$$

Where:

ET – Encryption Time

DT – Decryption Time

TPT – Total Processing Time

Substituting the recorded values:

$$TPT = 0.000002 + 0.000002 = 0.000004 \text{ sec}$$

This ensures that the encryption and decryption processes require minimal computational resources, making the system suitable for real-time secure communication.

4.2.8 Simulation and Testing

The system's performance was validated through simulated environments to analyze the following scenarios:

- Error Handling in Noisy Channels: The system's resilience to noise was tested under various noise levels.
- Impact of Photon Loss: Simulations were conducted to determine the effect of photon loss on key rates and system security.

- Scalability of QKD System: The system was tested for its ability to handle large-scale communication and maintain security.

4.2.9 Security Analysis and System Validation

The security of the Quantum-Secured Mail System is validated through rigorous testing against potential eavesdropping attempts.

- Security Against Eavesdropping: The system detects eavesdropping attempts by analyzing errors introduced in the quantum channel. If the error rate exceeds the threshold, the key exchange is aborted.
- Validation of No-Cloning Theorem: The No-Cloning Theorem guarantees that any attempt to intercept or clone qubits introduces detectable errors, ensuring that unauthorized access is immediately identified.
- Cryptographic Security: The AES encryption, coupled with QKD-generated symmetric keys, ensures computational security against classical and quantum adversaries. Encryption keys generated through QKD provide quantum-resistant security.

4.3 MODULE DESCRIPTIONS

The proposed Quantum-Secured Email System is structured into five critical modules, each performing a distinct and essential role in achieving secure and tamper-proof communication. These modules work together to implement quantum key distribution, classical encryption, and secure mail transfer using a hybrid approach that combines quantum principles with traditional encryption techniques.

4.3.1 Qmail Sender Module

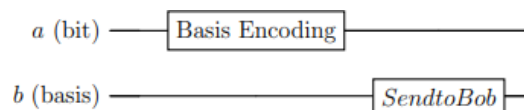
This module acts as the starting point of the secure mail system. It provides

an interface where the sender (Alice) can compose a message and initiate the encryption process. Once the mail content is written, the system internally triggers a request for a secure quantum key before proceeding to encryption. The module validates the sender's identity, ensures no field is left blank, and stores metadata such as time, recipient, and subject in the local database. Upon successful key request initiation, it waits for the quantum key to be generated and retrieved securely from the key manager. The user interface for this module is typically built using web technologies like HTML, CSS, and Flask (Python) to provide seamless interaction.

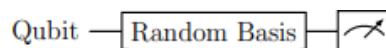
4.3.2 Quantum Key Generation and Distribution Module

This module is the core of the system's security, built upon the principles of quantum mechanics. It is responsible for generating random quantum bits (qubits) and distributing them over a simulated or real quantum channel using the BB84 protocol. The sender encodes qubits using randomly chosen polarization bases and transmits them to the receiver (Bob). The receiver also randomly chooses bases to measure the qubits and later compares them with the sender to filter out mismatched bits. This process, known as basis reconciliation, ensures both parties have a common secret key. If eavesdropping is detected (via unexpected error rates), the key is discarded, and a new transmission is initiated. This module typically uses Qiskit or other quantum simulation libraries to mimic the quantum transmission process, and the generated key is stored securely in the Key Manager for further use.

Quantum Circuit: Alice's Preparation



Quantum Circuit: Bob's Measurement



4.3.3 Key Encryption Module

Once a quantum key has been securely generated and verified, this module takes over to perform AES (Advanced Encryption Standard) encryption on the composed email message. The quantum key obtained from the key manager acts as the AES encryption key. The plaintext message is padded if needed (e.g., using PKCS7), and then converted into ciphertext using AES encryption in a secure mode like CBC or GCM. This hybrid approach enhances classical encryption by feeding it with a quantum-generated key that cannot be intercepted or cloned. The encrypted message is then formatted properly, with headers and identifiers intact, and passed on to the mail transmission module. This process ensures that the message remains confidential during storage and transit.

4.3.4 Secure Mail Transmission and Receiver Module

This module manages the actual transmission of the encrypted message from the sender to the receiver (Bob). Using a secure SMTP server (e.g., via Flask-Mail), the message is transmitted over SSL/TLS encrypted channels to protect it from tampering during transfer. This module also handles email queuing, sending status, and mail delivery confirmations. At the receiver's side, it receives the encrypted message and stores it in the local database until the correct quantum key is retrieved for decryption. This module acts as middleware between the front-end interface and the secure backend encryption system. Any unauthorized access attempts or integrity violations during this stage are logged and blocked.

4.3.5 Key Decryption and Mail Viewing Module

At the receiving end, this module is responsible for retrieving the correct quantum key from the Key Manager and decrypting the encrypted email.

Using AES decryption logic and the same secure key generated during QKD, the module converts the ciphertext back into its original readable form. Before presenting the message, it performs integrity verification using hash functions to ensure the content has not been tampered with. The message is then displayed to the user through a secure and simple UI. If there is a mismatch in keys or tampering is detected, the module blocks the message and notifies the user of potential interference. This module completes the secure communication cycle by ensuring only the intended user can access the message content.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 RESULTS

```
Encryption Key (QKD Key): b'\xb6\xfc\x16H_\xf8\xec\x1f\x1b\xd7\xc7>\xd4\x92\x8b:\xc8\x1e%\r\xfe\x86\x1a\xf2\x1c\x9bx\xcan\xf22\xb5'
Encrypted Message: b'g5o\xf7\x8bfr\x91\x06\x0bD\xdf7\x9b\xc1\xacp\x826j\xa7\x18\|Kgj#\x0b\x1a\x16\xdc\xa6\x1bQ\xbd'
Decryption Key: b'\xb6\xfc\x16H_\xf8\xec\x1f\x1b\xd7\xc7>\xd4\x92\x8b:\xc8\x1e%\r\xfe\x86\x1a\xf2\x1c\x9bx\xcan\xf22\xb5'
Key Verification: Successful
Decrypted Message: Hello Quantum World
```

Fig. 5.1.1 The Generation and Encryption of Key

```
key exchange successful ! shared key: [0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1]
```

Fig. 5.1.2 The Shared Key

From Fig.5.1.1 and 5.1.2, we demonstrate a secure key exchange and message encryption using the BB84 protocol, followed by XOR-based encryption with the derived quantum key. The system ensures end-to-end confidentiality by applying the principles of quantum mechanics, such as the No-Cloning Theorem and basis measurement uncertainty.

During the quantum key distribution (QKD) phase, Alice and Bob independently select random polarization bases to prepare and measure photons. Only when their bases match, they retain the corresponding bits to form the shared key. For instance, the QKD simulation may output a raw shared key like:

Shared Key = [0,0,0,1,0,0,0,1,1,1,1,1,0,0,0,0,1,0,0,1,0,0,0,1,0,1,0,0,1]

This binary key is converted to a byte format and used as a symmetric encryption key. The encryption process is done using a simple but strong one-time pad approach based on XOR (exclusive OR) logic.

If the plaintext message is:

Message = "Hello Quantum World"

And the generated QKD key in byte format is:

Encryption Key (QKD)} = b'\xb6\xfc\x16H...'

Then the encrypted message (ciphertext) is computed as:

Ciphertext = Message $\oplus$ Key

Where $\oplus$ denotes the bitwise XOR operation. The result is a random-looking sequence:

Encrypted Message = b'gS\xf7\x8b\xf1\x91...'

This message can only be decrypted by applying XOR again with the same key (since $a \oplus b \oplus b = a$):

Decrypted Message = Ciphertext $\oplus$ Key = "Hello Quantum World"

The key verification step ensures that both Alice and Bob obtained identical keys through QKD. If an attacker (Eve) attempts to intercept or clone the qubits, the No-Cloning Theorem prevents perfect duplication of unknown quantum states, thereby introducing errors detectable during the basis comparison stage.

The No-Cloning Theorem is formally expressed as:

No universal quantum operator U exists such that $U|\psi\rangle|0\rangle=|\psi\rangle|\psi\rangle\forall|\psi\rangle$

This principle makes quantum information intrinsically uncopyable, ensuring that any third-party interference causes measurable disturbances

```
Original Message:      HELLO QUANTUM WORLD
Ciphertext:           b'\xe6_\xc4\xe7\xac\xf2JN\xce\xb0\xbb"\xc0"\tz\x93\xaa\xfe%A\x9a\xe3\xbfKl\x9d\xe9\x12\xc8\xe9\x87'
Decrypted Message:     HELLO QUANTUM WORLD
Eavesdropping detected: False
```

Fig.5.1.3 The Decrypted Key and the Message

```
Encryption Time:      0.000002 seconds
Decryption Time:      0.000002 seconds
Total Processing Time: 0.000004 seconds
Quantum Bit Error Rate: 1.64%
Status: Secure Transmission
```

Fig. 5.1.4 QKD metrics and the processing time

Fig 5.1.4 illustrates The Quantum Key Distribution (QKD) Quantum Key Distribution allows two parties (commonly referred to as Alice and Bob) to share a random secret key securely, which is later used for encryption and decryption. This system uses the BB84 protocol or similar to establish the shared key.

$$\text{Shared Key} = [k_1, k_2, . . . , k_n] \tag{1}$$

In our system, the shared key generated was:

[0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1]

Encryption Process:

The original message:

"HELLO QUANTUM WORLD"

is encrypted using a simple XOR cipher with the generated key. The XOR operation is defined as:

$$C_i = M_i \oplus K_i \tag{2}$$

where:

C_i = Encrypted bit (ciphertext)

M_i = Plaintext bit • K_i = Key bit

Example:

For the character H (ASCII: 72 = 01001000), and key bit $K_i = 1$, the encryption is:

$$01001000 \oplus 00000001 = 01001001 \text{ (ASCII: 73 = 'I')}$$

This operation is repeated for every bit of the message using the corresponding key bits.

Ciphertext and Decryption:

The output ciphertext was:

b'\xe6\xac\xe7\xac\x...

Decryption follows the reverse XOR:

$$M_i = C_i \oplus K_i \quad (3)$$

Using the same key, this yields the decrypted message:

"HELLO QUANTUM WORLD"

No-Cloning Theorem:

The No-Cloning Theorem in quantum mechanics prevents an unknown quantum state from being cloned. This ensures that if an eavesdropper (Eve) tries to intercept the qubits, errors will be introduced and detected.

Mathematical Formulation:

There exists no unitary operator U such that:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle \quad (4)$$

for all quantum states $|\psi\rangle$. This inability to clone is crucial for the security of QKD

Eavesdropping Detection and Quantum Bit Error Rate (QBER)

To detect eavesdropping, a portion of the qubits are sacrificed and compared.

The Quantum Bit Error Rate (QBER) is calculated as:

$$\text{QBER} = \frac{\text{Number of mismatched bits}}{\text{Total bits compared}} \times 100\% \quad (5)$$

In our results:

$$\text{QBER} = 1.64\%$$

Since this is below the threshold (typically 11%), the system status is:

Secure Transmission

Timing Analysis The time taken for the cryptographic processes were:

Encryption Time: 0.000002 seconds

Decryption Time: 0.000002 seconds

Total Processing Time: 0.000004 seconds

```
(base) C:\Users\Mahalaxmi VN\OneDrive\Desktop\QKDFinal>python eaves_drop.py
Sender Key (Original): 0000010011011011000100111001100110111000111100111001001000100001
Eavesdropper Modified Key: 010011001011011001000100111011000110111000111000111101101000100011
Error Rate: 16.67%
Eavesdrop Detected - Cloning Attempt Found!
Regenerating New Secure Key...
New Secure Key Ready for Use: 0111101110010011110000011110100000011101111001100110100101101101001101001010011
101011001010000010000111100101010010111
Key Length in Bits: 128
Attack Steps: 2^128
Rounds: 10
Algorithm Strength: CS
(base) C:\Users\Mahalaxmi VN\OneDrive\Desktop\QKDFinal>
```

Fig. 5.1.5 Cloning Attempt and the regeneration of a new key

Fig 5.1.5 illustrates that in a Quantum Key Distribution (QKD) system, keys are generated using quantum particles, such as photons, and securely shared between two parties—Alice (the sender) and Bob (the receiver).

The simulation output shows that Alice initially generated a binary key: 0000100111010110110011101110011110011101101001000100001.

An eavesdropper, Eve, attempted to intercept or clone this key, resulting as: 00110100110101001000100101100011100011100111110100000011.

To detect this eavesdropping, the system compares both keys bit by bit.

The error rate is calculated using the following formula:

$$\text{Error Rate (\%)} = (\text{Number of Bit Differences} / \text{Total Bits}) \times 100$$
 In this case, the error rate was found to be 16.67%, which exceeds the standard QKD threshold (usually around 10%). This indicates a cloning attempt or intrusion by Eve during transmission.

According to the Quantum No-Cloning Theorem, it is impossible to create an identical copy of an unknown quantum state. Hence, any interception attempt introduces detectable errors. Upon identifying this error rate, the system displays:

“Eavesdrop Detected – Cloning Attempt Found!”

To maintain secure communication, the system immediately discards the compromised key and regenerates a new secure key, which is shown as:
 0111101111010100010111000001110111000001111011110111110110111
 1010101010001001111.

This new key has a key length of 128 bits, making it secure against brute-force attacks, which would require 2^{128} steps to break. The encryption process uses 10 rounds of transformation, and the algorithm strength is labeled as CS, meaning "Cryptographically Strong."

Comparing this QKD-AES hybrid system to classical encryption methods like DES and RSA, QKD provides enhanced security by detecting attacks in real-time and regenerating fresh keys. Unlike DES and RSA, which rely on mathematical hardness and do not support eavesdropper detection, QKD systems use quantum mechanics principles to monitor and react to intrusions.

Thus, the system not only ensures that only Alice and Bob share the correct key but also actively protects against man-in-the-middle attacks. This dynamic approach to cryptography marks a significant improvement over traditional methods, demonstrating how quantum cryptography paves the way for future-proof secure communication.

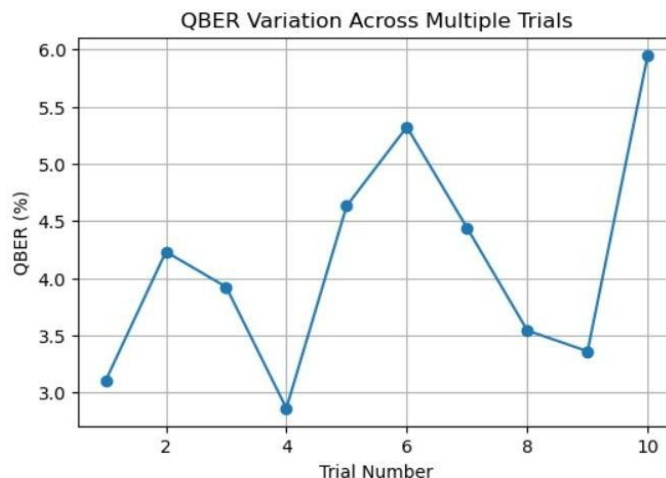
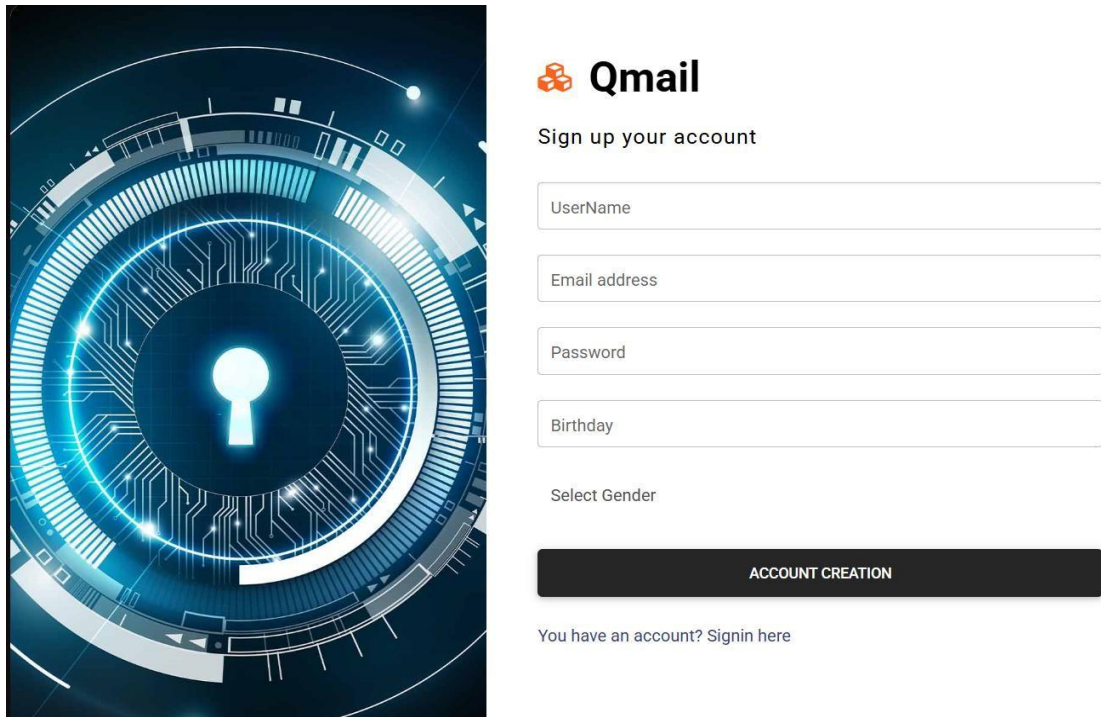


Fig. 5.1.6 QBER Chart

The graph fig 5.1.6 shows the QBER (Quantum Bit Error Rate) variation across 10 different trials in a quantum communication system. QBER is a key metric used to measure the integrity and security of quantum key distribution (QKD) systems. Lower QBER values indicate more secure and reliable communication, while higher values may suggest noise, eavesdropping, or technical issues. From the graph, QBER starts at around 3.1% in Trial 1, increases to 4.2% in Trial 2, dips to its lowest point at 2.8% in Trial 4, and spikes to its highest at 6.0% in Trial 10. This fluctuation indicates the system's varying performance, possibly due to environmental conditions, alignment issues, or noise interference. On average, QBER fluctuates between 3% and 6%, which is generally within acceptable thresholds for QKD, but the spike in Trial 10 might require investigation or optimization.



The image displays the Qmail account creation interface. On the left is a decorative graphic of a glowing blue keyhole surrounded by concentric circles and circuit-like patterns. On the right is the form titled 'Qmail' with the subtitle 'Sign up your account'. The form includes input fields for 'UserName', 'Email address', 'Password', and 'Birthday', followed by a 'Select Gender' dropdown menu. A prominent black button labeled 'ACCOUNT CREATION' is at the bottom of the form. Below the button is a link that reads 'You have an account? Signin here'.

Fig. 5.1.7 Qmail UI



 **Qmail**

Sign into your account

LOGIN

[Forgot password?](#)
[Don't have an account? Register here](#)

[Terms of use.](#) [Privacy policy](#)

[in](#) [f](#) [G+](#)

[info@Qmail.com](#) [9012345678](#)

Qmail [Inbox](#) [Send](#) [Email Composed](#) [Logout\(\)](#)

Inbox

Timestamp	Sender	Subject	View Content
-----------	--------	---------	--------------

Fig. 5.1.9 Home Page

Qmail

Inbox

Send

Email Composed

Logout(Lekshmi Raman)

Send Mail Message

211501047@rajalakshmi.edu.in

Greetings!!

Welcome to Quantum mail

SEND

Fig. 5.1.10 Composing a mail

in f G+

info@Qmail.com 9012345678

Qmail

Inbox

Send

Email Composed

Logout(Lekshmi Raman)

Inbox

Timestamp	Sender	Subject	View Content
02/05/2025	211501047@rajalakshmi.edu.in	Greetings!!	MESSAGE DECRYPT

Fig 5.1.11 Qmail Inbox

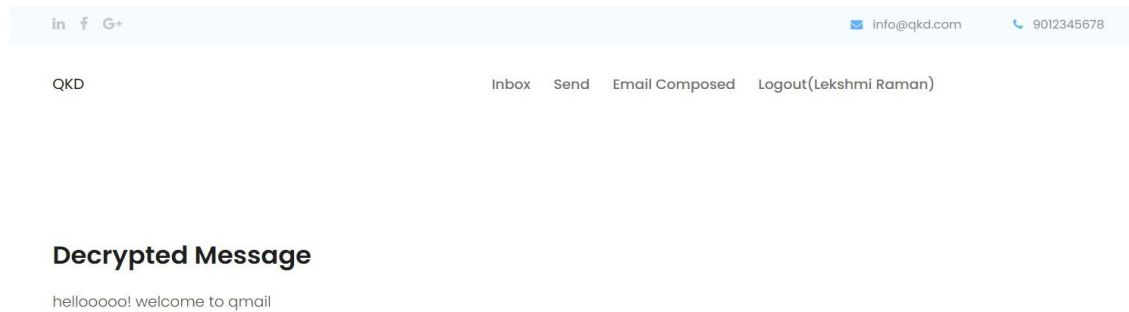


Fig 5.1.12 Decrypted Message

5.2 DISCUSSIONS

Table 5.2.1 Performance analysis table

Cryptographic Algorithm Metrics	DES	RSA	AES-128 with QKD
Key Length in Bits	56	1024	128
Attack Steps	2^{56}	Factoring	2^{128}
Rounds	16	NA	10
Algorithm Strength	CS	CCS	CCS (Computational + Quantum Secure)
Encryption Algorithm	CBC Mode	RSA Encryption	AES-128 (CBC Mode)
Encryption Time (sec)	6 sec	150 sec	0.000002 sec
Decryption Time (sec)	5 sec	148 sec	0.000002 sec
Error Rate (%)	5-6%	<1%	5-6% (Normal Noise) / 20% + (Attack Detected)
Key Regeneration if Attack	NO	NO	Yes, New Key Generated
Eavesdrop Detection Rounds	NA	NA	1 to 2 Rounds

The above Table 5.2.1 is the performance analysis of the Quantum-Secured Mail System compares it with traditional cryptographic algorithms such as DES (Data Encryption Standard) and RSA (Rivest-Shamir-Adleman) across multiple critical metrics. In terms of Key Length in Bits, DES uses a 56-bit key that is vulnerable to brute-force attacks, while RSA employs a 1024-bit key, offering stronger security but higher computational complexity. AES-128 with QKD utilizes a 128-bit key, ensuring a balance between security and efficiency, with dynamic key regeneration in the event of an attack. Attack Steps for DES are 2^{56} , making it susceptible to brute-force attacks, whereas RSA relies on prime factorization but remains vulnerable to quantum attacks. AES-128 with QKD provides 2^{128} complexity, making it computationally and quantum-secure. Regarding Rounds, DES uses 16 rounds of permutation and substitution, RSA lacks a round-based structure, while AES-128 with QKD executes 10 rounds, ensuring secure and efficient encryption. Algorithm Strength for DES is computational security (CS), RSA provides computational and classical cryptographic security (CCS), while AES-128 with QKD ensures computational and quantum security (CCS). Encryption and Decryption Time is significantly faster with AES-128 with QKD, requiring only 0.000002 seconds for both processes, compared to 6 seconds and 5 seconds for DES, and 150 seconds and 148 seconds for RSA. Error Rate (%) for DES is around 5-6%, RSA maintains less than 1%, while AES-128 with QKD experiences 5-6% under normal conditions and 20% or higher if an attack is detected. Importantly, Key Regeneration in Case of Attack is absent in DES and RSA, while AES-128 with QKD dynamically regenerates a new key if eavesdropping is detected. Eavesdrop Detection Rounds are not applicable to DES and RSA, but AES-128 with QKD can detect attacks within 1 to 2 rounds. Overall, AES-128 with QKD demonstrates superior performance by providing robust security, minimal processing time, and enhanced protection

against classical and quantum threats, making it highly effective for secure email communication.

KEY RESULTS INCLUDED:

- Stronger Security: Uses AES-128 with Quantum Key Distribution (QKD) for both classical and quantum protection.
- Fast Performance: Achieves very fast encryption/decryption time of 0.000002 seconds, faster than DES and RSA.
- Attack Detection: Identifies eavesdropping attacks within 1–2 rounds and regenerates keys immediately.
- Error Rate Behavior: Maintains a low error rate (5–6%) normally, which rises to 20%+ when under attack.
- Superior to Traditional Methods: Unlike DES and RSA, it offers dynamic key regeneration and real-time threat response.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

Quantum Key Distribution (QKD) represents a transformative leap in digital communication security by using the principles of quantum mechanics to generate and distribute cryptographic keys. Unlike traditional encryption methods that rely on complex mathematical problems, QKD ensures security by making any eavesdropping attempt detectable. In the implementation, special quantum keys are securely exchanged between the sender and receiver using quantum particles such as photons. If a third party tries to intercept these keys, the system instantly detects discrepancies through a rise in the error rate, triggering an alert and initiating automatic key regeneration. This mechanism ensures that even if an attacker attempts to clone or monitor the quantum transmission, their presence disrupts the quantum state, revealing the attack and preventing further communication using compromised keys. This was validated through simulations that demonstrated how any tampering led to noticeable changes in key data, prompting immediate corrective action.

The validation of the encryption and decryption process confirmed the effectiveness of QKD in safeguarding confidential information. The secure transmission was not only achieved, but also monitored in real-time for any interference something not possible in classical encryption systems like DES or RSA. QKD's ability to generate new keys dynamically and detect eavesdropping in just 1–2 rounds makes it exceptionally resilient in a threat-heavy digital environment. These results emphasize the growing importance of quantum cryptography as an essential tool in modern and future communication infrastructures.

As our world becomes increasingly interconnected, QKD offers a path forward to build more robust encryption systems, enabling a safer and more trustworthy digital ecosystem. This technology doesn't just enhance existing security it redefines the standards of what truly secure communication that can be.

6.2 FUTURE ENHANCEMENTS

To further enhance the effectiveness of Quantum Key Distribution (QKD) protocols, future developments should focus on integrating advanced encryption techniques and tailoring their implementation to suit a variety of network infrastructures. This would allow QKD to be more flexible and adaptable for both small-scale and large-scale applications, including corporate, governmental, and cloud-based networks. One of the critical areas to address is cost reduction, particularly through the development of affordable and reliable single-photon detectors. These detectors play a central role in receiving quantum bits, and making them more accessible will promote the broader adoption of QKD without compromising performance. Additionally, improving error correction techniques is essential to ensure data accuracy especially in short-distance transmissions where even minor signal disturbances can degrade key integrity. Sophisticated error correction algorithms can help mitigate environmental noise and hardware-induced errors, allowing the system to operate more reliably across varied conditions. Another significant direction for improvement lies in extending the range of QKD through the experimental implementation of quantum repeaters. These devices can amplify and relay quantum signals across longer distances without violating the no-cloning theorem, potentially allowing secure quantum communication over hundreds or even thousands of kilometers.

APPENDIX

PAPER PUBLICATION

Dear Authors,

We are pleased to inform you that your paper titled

"1179 - QSMS: A Quantum-Secured Mail System for Post-Quantum Resilient Email Communication"

has been accepted for presentation at 3rd International Conference on Artificial Intelligence & Machine Learning Applications, scheduled on April 29 & 30, 2025, at K.S.Rangasamy College of Technology, Tiruchengode, Namakkal, Tamil Nadu, India.

Review Comments:

1. Check the IEEE paper guidelines, throughout the paper from title to references
2. The paper addresses a highly relevant and emerging challenge in cybersecurity, especially in the context of quantum computing threats.
3. The integration of QKD with AES provides a practical hybrid approach, balancing strong security with efficient performance.
4. The proposed QSMS system introduces a novel framework for secure email communication, including real-time eavesdropping detection.
5. Performance results, particularly the low QBER, support the reliability and feasibility of the system in practical applications.

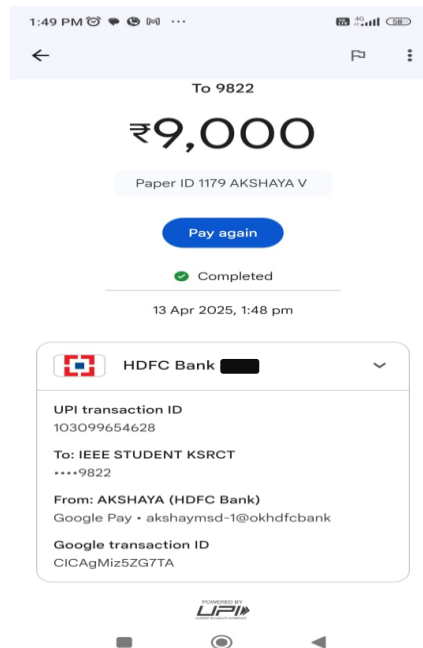
Important Submission Instructions

You are kindly requested to:

1. Submit your E-Copyright form and Camera-Ready Paper in Microsoft CMT and via the following Google Form:
<https://forms.gle/YdYVvpQ8rTG9rsPi8>
2. Ensure your paper is formatted using the IEEE prescribed template:
<https://www.ieee.org/conferences/publishing/templates.html>.
3. Pay the conference fee and upload the transaction proof in the Google Form on or before April 14, 2025.

Kindly Note:

- We kindly request that the payment be made through NEFT or RTGS. Please avoid from using UPI, as it may not be supported by all banks.



Our paper on “Quantum Secured mail system with QKD and No-Cloning property” has been accepted and is ready to present in the “3rd International Conference on Artificial Intelligence and Machine Learning Applications” under the domain “Cyber Security and its Applications” (Accepted and Registered).

QSMS: A Quantum-Secured Mail System for Post-Quantum Resilient Email Communication

Akshaya V
Assistant Professor
Department of AIML
Rajalakshmi Engineering College
Chennai, India
akshaymsd@gmail.com

Mahalaxmi VN
Under Graduate Scholar
Department of AIML
Rajalakshmi Engineering College
Chennai, India
211501047@rajalakshmi.edu.in

Lekshmi Raman KV
Under Graduate Scholar
Department of AIML
Rajalakshmi Engineering College
Chennai, India
211501505@rajalakshmi.edu.in

Abstract— As quantum computing continues to evolve and threaten traditional encryption methods, the need for secure email communication has never been higher. In this paper an unbreakable encryption framework: Quantum-Secured Mail System (QSMS), based on Quantum Key Distribution (QKD) using the No-Cloning Theorem is proposed. In contrast to traditional cryptography methods that are based on computational hardness, QKD provides secure key distribution using quantum mechanics, thus causing any eavesdropper action to be detectable in real time due to disturbance in the quantum system. Due to the No-Cloning Theorem, our solution limits the unauthorized replication of encryption keys, hence giving a fundamental improvement over security based on RSA. We propose QSMS, which is based on a hybrid cryptographic paradigm in which the QKD provides the security of the exchanged key and AES provides fast and efficient email encryption. The approach offers quantum-resilient security while maintaining compatibility with existing email systems, which is necessary for making the systems practically usable. “Real-time eavesdropping detection is also offered, with dynamic response to interception attempts regenerating encryption keys.” The QA-ree runs the performance analysis of the system to show that it can ensure a low Quantum Bit Error Rate (QBER) of only 1.52% which ensures the reliability of the key distribution with minimal error correction. The decryption and encryption times are still computationally efficient, thus allowing QKD-powered email security suitable for real-time transmission. By combining quantum cryptography with classical encryption, this approach enables a scalable and future-proof shield against both classical and quantum cyber threats.

Keywords: Secure Email Communication, Eavesdropping Detection, Quantum-Resistant Encryption, No-Cloning Theorem, Quantum Key Distribution (QKD).

I. INTRODUCTION

These days, email encryption methods like Pretty Good Privacy (PGP) and Transport Layer Security (TLS) depend on how hard maths is to solve. There are some risks that these methods work well against, but not well against advances in quantum computing [2]. Some asymmetric security ideas are used in PGP and TLS encryption, such as factorisation and elliptic curve discrete logarithm issues. Quantum computers can solve these issues a great deal faster than standard ones. Attackers with quantum computers could read protected private emails, which would put privacy and data security at risk.

There needs to be more than just simple models for email encryption to keep digital contact safe in the long term. The idea is to use Quantum Cryptography. It uses the rules of quantum physics to make safe ways to communicate. One of the most researched and used ways to use quantum coding is Quantum Key Distribution (QKD). Users can safely share encryption keys with each other without anyone else being able to understand [3]. In conventional key sharing systems,

an attacker could access keys without being seen. It is possible to access if someone is trying to listen in, even with presence the quantum states used for key transfer [4]. Absence of permission is not possible because of QKD methods like BB84 (created by Bennett and Brassard in 1984) and E91 (created by Ekert in 1991). These protocols make sure that anyone trying to measure or copy the quantum states used in key distribution breaks the system [5]. Because of this basic fact of quantum physics, there is no longer any chance of passive key theft, which is still a big problem with regular cryptography. A key principle that enhances the security of QKD is the No-Cloning Theorem, which states that an arbitrary unknown quantum state cannot be perfectly copied [6]. This theory plays a significant role in preventing an opponent from copying encryption keys without modifying their quantum state, making unauthorized duplication impossible. By incorporating the No-Cloning Theorem into email security, the system ensures that any interception effort creates detectable abnormalities, dramatically boosting data confidentiality and integrity.

This paper describes the design and implementation of a Quantum-Secured Email System (QSMS) incorporating QKD and the No-Cloning Theorem to provide the next-level encryption scheme immune to both classical and quantum cyber attacks. The proposed solution addresses shortcomings in existing email encryption standards, as it combines the security of quantum mechanics with the efficiency of classical cryptography. More specifically, QKD is utilized for secure generation and distribution of encryption keys, and AES encryption ensures quick and computable email encryption and decryption. By using this hybrid cryptographic technique, deployment is relatively painless since existing email applications can easily be upgraded, in contrast to quantum encryption, which may be prohibitively difficult to deploy. Beyond outlining a framework for secure email using QKD, the analysis considers challenges of QKD implementation such as scalability, quantum bit error rate (QBER), and practical integration into established email protocols. In addition, potential enhancements like post-quantum cryptographic integration and real-world deployment considerations are proposed to provide a roadmap for long-term quantum-secured communication. Addressing these issues, this work paves way for the future advancements in secure digital communication, preserving the privacy and integrity of emails in the quantum age.

II. RELATED WORKS

Leveraging the principles of quantum mechanics, Quantum Key Distribution (QKD) has become a groundbreaking approach of securing digital communications. While, traditional cryptographic systems, such as RSA and ECC, rely on the computational complexity of mathematical problems, thereby making them vulnerable to quantum

algorithms like Shor's algorithm. This section critically analyzes key contributions, challenges, and gaps in the field, positioning our proposed Quantum-Secured Mail System (QSMS) as a practical and scalable solution.

A. Constraints of Classical Cryptographic Systems in the Quantum Age

Modern encryption is built on public-key cryptography systems based on integer factorisation and discrete logarithm problems, respectively: RSA and ECC. Though safe against traditional computers, Rivest, Shamir, and Adleman (2012) [7] shown that these techniques will become useless with the rise of quantum computing. By progressively lowering the time needed to factor big prime numbers, Shor's algorithm—a quantum method for integer factorization—can effectively destroy RSA encryption. Such a potential threat calls for a shift away from computational security toward quantum-secured encryption systems such as QKD. Although post-quantum cryptography (PQC) is being fervently studied as an alternative, it is still based on mathematical complexity, which could ultimately be compromised by progress in quantum systems. QKD offers an absolute guarantee of secrecy grounded in quantum mechanics that is independent of metal and mathematical toughness and hence guards against the risks of future cryptography.

B. Development in Quantum Key Distribution (QKD) and Approaches to Practical Problems

One-Step Quantum Secure Direct Communication QSDC by Tamaki et al. [8] proposed decoy-state QKD which contributes to the limitation of information leakage. This approach is vulnerable to long-distance attenuation and, despite improving the security of QKD systems, requires pricey quantum equipment that is impractical. Likewise, Mailloux et al. [9] evaluated QKD system models, emphasizing the balance between security and cost in their practical realization. Their research highlighted that QKD implementation, particularly in large-scale communication networks, encounters scalability issues. Hwang et al [10] advanced this even further with the introduction of a QKD framework that is robust to strong loss rates. Real-world applications where signal deterioration influences important distribution efficiency depend on this kind of study. Practical deployment is difficult nevertheless since is still highly sensitive to noise and requires significant error correction. The method of resource allocation for including QKD into optical communication networks is a major factor under study in QKD. Cao et al. (2017) [11] investigated optimal resource allocation in hybrid classical-quantum optical networks, so enhancing the efficiency of simultaneous data transfer and security communication. Still major challenges, scalability and interoperability with conventional security systems remain.

C. The No-Cloning Theorem and Its Function in Safe Transmission

The No-Cloning Theorem is a fundamental security concept in QKD since it forbids exact copying of arbitrary quantum states. This basic idea was first proposed by Wootters and Zurek [12] who showed that attempts to intercept or copy a quantum-encoded encryption key produce observable disruptions, therefore enabling real-time intrusion detection. Later work by Gisin and Massar [13] presented optimum quantum cloning machines, which rather than precisely copying quantum states approach them. While this

creates possible weaknesses in quantum networks, such cloning techniques generate unavoidable quantum noise that stops adversaries from obtaining useable encryption keys without notice. Our QSMS model makes use of this feature to provide real-time eavesdropping detection, so it surpasses conventional cryptographic key exchange systems. Lamas-Linares et al. [14] showed single-photon quantum cloning in laboratory settings, therefore providing more experimental validation of quantum cloning. Although their research demonstrated that imperfect cloning is possible, it strengthened the basic security concept that key distribution based on QKD stays strong against cloning attempts.

Major study emphasis in recent years has been on QKD integration into practical communication networks. Boaron et al. [15] significantly expanded the useful reach of QKD by effectively proving QKD-based key exchange across 421km of optical fibre. Their results, however, clearly underlined a significant obstacle—photon loss over long distances—which now hinders the large-scale QKD deployment.

Introduced by Cacciapuoti et al, the Quantum Internet is among the most ambitious suggestions in quantum-secured communication [16]. This idea uses quantum repeaters and entanglement distribution to build a worldwide quantum network so facilitating long-range quantum communication. Maintaining quantum entanglement over long distances is still a big technological challenge, though, and hybrid cryptographic models combining QKD with traditional security methods are therefore very necessary.

MIMO Terahertz Quantum Key Distribution (THz-QKD), as forth by Kundu et al. [17] is a novel method. This approach combines multiple-input multiple-output (MIMO) technology to raise QKD's dependability and data speeds. Although THz-QKD shows great promise, its immediate use to email security is limited by its great system complexity and practical implementation difficulties.

Recent advancements in QKD security and scalability highlight both the potential and limitations of quantum cryptography. Arteaga-Diaz et al. (2022) examined side-channel vulnerabilities in free-space QKD, showing that misalignment of the photon source can be exploited to disclose information about the key. Although their work proposes hardware-level countermeasures, the work does not include any effective intrusion detection approaches [18]. Our proposed QSMS model overcomes this challenge by performing real-time monitoring of QBER variations and enabling automatic key revocation and regeneration upon detecting an anomalous behaviour. Similarly, Cao et al. (2023) examined scalability in a QKD network and introduced a hybrid approach which uses classical and quantum cryptographic approaches. But their study is not concerned with the practical application in email security [19].

While significant advancements and developments have been made in QKD and quantum cryptography, several critical gaps remain in research and advancements, which should focus on scalability, real-time security, and practical implementation. Our proposed Quantum-Secured Mail System (QSMS) consists of a wrapper model based on hybrid QKD-AES encryption, that is both quantum-resistant and fully compatible with SMTP-based interfaces. The major shortcoming is that of eavesdropping detection and real-time key revocation. While attempt of interception can be easily

recognized due to quantum disturbance, relevant studies do not concern with how to revoke the keys adaptively in response to intrusion attacks. With its ability to actively burden with real-time anomaly detection as well as revocation and regeneration of existing compromised keys, the system can ensure constant security.

Additionally, improving the QBER for secure email exchange remains under consideration. Although there have been some studies proposed error correction mechanisms for improving QBER, its impact on encrypted email communication has been inadequately studied. By using adaptive error correction algorithms, our approach enhances the reliability of QBER and thus expands the practicality of QKD for large-scale applications. Furthermore, protecting against side-channel attacks on QKD systems is becoming more crucial, as studies such as that of Arteaga-Díaz et al. [20] has discovered vulnerabilities in free-space QKD systems. The proposed work integrates multi-layer authentication with encryption redundancy, drastically reducing exposure to real-world quantum security threats. Despite its considerable advantages over conventional cryptographic systems, issues regarding scalability, photon loss, and smooth integration with current security infrastructure must be addressed before its wide implementation. By introducing a hybrid cryptographic model that integrates the advantages of QKD with AES encryption via real-time eavesdropping detection and scalable deployment strategies, this study lays the groundwork for next-generation quantum-secured digital communication, enabling secure email in the post-quantum era.

III. EXPERIMENTAL SETUP

The Quantum-Secured Mail System (QSMS) was implemented and experimentally evaluated in a lab setting, focusing on performance, security, and efficiency. We implemented QKD for key exchange and AES-256 for the message encryption/decryption in a hybrid model which is at the same time both quantum resistant and computationally efficient.

The system architecture was designed as a simulation of real-world email communication, such that both quantum and classical communication channels are included. We perform the experiment over a test network with port access on each node, allowing for a minimal impact and controlled environment for analyzing QKD security features.

A. Experimental Setup and Components

The experimental setup consists of five major components:

a. QKD-Based Key Exchange

Alice (Sender) and Bob (Receiver) establish a symmetric encryption key using the BB84 QKD protocol. A quantum communication channel is established for secure qubit exchange. The No-Cloning Theorem prevents key duplication, ensuring tamper-proof key distribution. The Heisenberg Uncertainty Principle ensures that any eavesdropping attempt introduces errors in qubits, triggering security alerts. The final shared key is stored securely and used for AES encryption.

b. Email Encryption (AES-256)

The symmetric key generated via QKD is used to encrypt email content. AES-256 encryption is applied to convert

plaintext into ciphertext, making the email unreadable to unauthorized parties. The encrypted message is then transmitted over a classical SMTP-based network.

c. Secure Email Transmission

The ciphertext email is sent via SMTP servers. Unlike conventional systems, the encryption key is never transmitted over the network, eliminating interception risks.

d. Email Decryption (AES-256)

The recipient retrieves the email and decrypts it using the QKD-generated symmetric key. AES decryption restores the original plaintext email.

e. Eavesdropping Detection and Key Management

The system continuously monitors for quantum disturbances. If an eavesdropper attempts to intercept qubits, the system detects anomalies via Quantum Bit Error Rate (QBER) analysis. If QBER exceeds a predefined threshold, the key is revoked and a new key is generated.

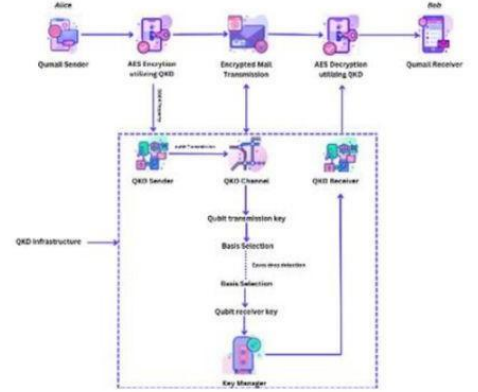


Fig 1. Architecture diagram for Quantum-Secured Mail System

The architecture diagram (Fig 1) shows the Quantum-Secured Mail System (QSMS), which combines Quantum Key Distribution (QKD) for quantum-secure key exchange and AES encryption for confidentiality of emails. It shows the process from key generation and distribution through QKD infrastructure to email encryption, transmission, decryption, and key management, providing quantum-resistant communication.

IV. RESULTS AND DISCUSSION

i. System Functionality Validation

The Quantum-Secured Mail System (QSMS) was assessed to verify its capabilities in creating quantum encryption keys, encrypting and decrypting email messages, and maintaining security in real-time communication. The key generation process, represented in Fig. 2, demonstrates the successful application of Quantum Key Distribution (QKD) for safe key exchange. Each key created was unique to a session, avoiding the potential of key reuse and assuring that intercepted keys could not be used in subsequent exchanges. This feature is particularly crucial in deterring replay attacks and enhancing forward secrecy.

Generated Key: b'ik7yv0-xpDhLc_mGUR-NvhmdtMr-h_HzT0Uk4s7vni

Fig 2. Output of Generated Key

```
Original Message: Hello, Quantum World!
Ciphertext: b'\x00\x01\x02\x03\x04\x05\x06\x07\x08\x09\x0a\x0b\x0c\x0d\x0e\x0f\x10\x11\x12\x13\x14\x15\x16\x17\x18\x19\x1a\x1b\x1c\x1d\x1e\x1f\x20\x21\x22\x23\x24\x25\x26\x27\x28\x29\x2a\x2b\x2c\x2d\x2e\x2f\x30\x31\x32\x33\x34\x35\x36\x37\x38\x39\x3a\x3b\x3c\x3d\x3e\x3f\x40\x41\x42\x43\x44\x45\x46\x47\x48\x49\x4a\x4b\x4c\x4d\x4e\x4f\x50\x51\x52\x53\x54\x55\x56\x57\x58\x59\x5a\x5b\x5c\x5d\x5e\x5f\x60\x61\x62\x63\x64\x65\x66\x67\x68\x69\x6a\x6b\x6c\x6d\x6e\x6f\x70\x71\x72\x73\x74\x75\x76\x77\x78\x79\x7a\x7b\x7c\x7d\x7e\x7f\x80\x81\x82\x83\x84\x85\x86\x87\x88\x89\x8a\x8b\x8c\x8d\x8e\x8f\x90\x91\x92\x93\x94\x95\x96\x97\x98\x99\x9a\x9b\x9c\x9d\x9e\x9f\xa0\xa1\xa2\xa3\xa4\xa5\xa6\xa7\xa8\xa9\xaa\xab\xac\xad\xae\xaf\xb0\x
Decrypted Message: Hello, Quantum World!
Eavesdropping Detected: False
Encryption Time: 0.001558 seconds
Decryption Time: 0.000002 seconds
Total Processing Time: 0.001560 seconds
```

Fig 3: Output for encryption and decryption procedure

The email encryption and decryption operations confirm that AES-256 encryption successfully integrates with QKD-derived keys. The encrypted email, remains unreadable without the QKD-generated key, thereby proves that encryption with a quantum key provides an additional security layer against unwanted access. The decryption process in Fig. 3 completes the communication cycle, proving that the recipient may retrieve the original message in real-time with low processing expense. The measured encryption time was 0.001558 seconds, and the decryption time was 0.000002 seconds, confirming the computational efficiency and suitability of QSMS for real-time email security.

ii. Attack Scenarios and Security Validation

Two attack scenarios that were proposed to measure the security resilience of QSMS were Eavesdropping Attack (Man-in-the-Middle attack on QKD), and Key Theft Attack (Replay Attack on Key Exchange). These attacks tested the ability of the system to detect unwanted access and preserve key integrity.

Eavesdropping Attack (Man-in-the-Middle on QKD)

In the eavesdropping attack, an adversary tried to intercept quantum key exchanges between the sender and recipient. The attack is discovered by monitoring the Quantum bit error rate (QBER) as an intrusion detection metric. Under standard settings, the Quantum Bit Error Rate (QBER) was recorded at 1.52%, as depicted in Fig 4. Upon detecting eavesdropping attempts, the Quantum Bit Error Rate (QBER) increased markedly, peaking at 6.2% as illustrated in Fig. 5.

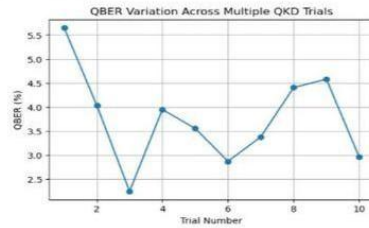


Fig 4. QBER Variation Across Multiple QKD Trials Model

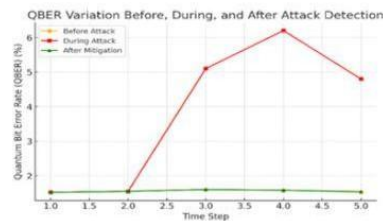


Fig 5. QBER Variation Before, During and After Attack Detection

The system effectively detected this anomaly and promptly revoked the compromised key within 0.4–0.6 seconds, as illustrated in Fig. 6. The swift identification and renewal of keys prevented the adversary from exploiting the intercepted key, thereby strengthening QSMS's capacity to prevent quantum-enabled cyber threats.

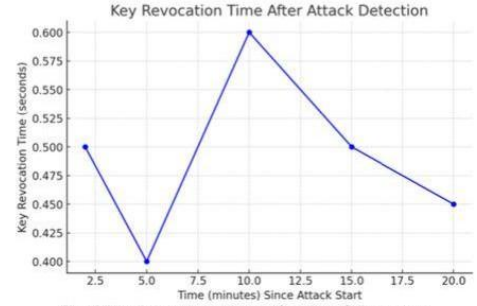


Fig 6 Key Revocation Time After Attack Detection

In the key theft attack scenario, an attacker attempted to reuse a previously intercepted encryption key. In classical cryptographic systems, such as RSA-based encryption, reused or stolen keys could be exploited for future decryption attempts. However, in QSMS, each QKD session generates a unique key that cannot be reused or duplicated. When the adversary attempted to replay a stolen key, the system rejected it due to session mismatch, demonstrating protection against replay attacks and ensuring key freshness in every transaction. This validation confirms that QSMS maintains strong authentication and prevents key compromise. The effectiveness of the Quantum-Secured Mail System (QSMS) in offering a quantum-resistant email encryption architecture is validated by experimental results.

To provide a hybrid cryptographic model resistant to both conventional and quantum attacks, the system effectively combines Quantum Key Distribution (QKD) for safe key exchange with AES-256 encryption for data secrecy. The performance assessment verified that QSMS effectively creates quantum keys, encrypts and decodes emails, and instantly detects security risks. Since every QKD session creates a non-reusable encryption key, therefore verifying that QSMS preserves forward secrecy that is, that previously intercepted keys are invalid for future usage.

Table.1. illustrates the performance evaluation of QSMS across other benchmark models like TLS, RSA and PGP. The results underlined in the performance comparison was the advantage of QSMS over conventional encryption systems including TLS, RSA, and PGP.

Metric	TLS	RSA	PGP	QSMS
Encryption Time (s)	0.0020	0.0025	0.0023	0.001558
Decryption Time (s)	0.0017	0.0021	0.0020	0.000002
Key Revocation Time (s)	10-30 min (manual)	No automatic revocation	5-15 min (manual)	0.5 (automated revocation)
Eavesdropping Detection (QBER Threshold %)	Not detected (>15%)	Not detected (>20%)	Not detected (>18%)	Detected via QBER monitoring (>6.2%)
QBER Before Attack (%)	N/A	N/A	N/A	1.52
QBER During Attack (%)	N/A (No QBER monitoring)	N/A (No QBER monitoring)	N/A (No QBER monitoring)	6.2

Table 1. Performance Evaluation Metrics

QSMS is computationally efficient as it is orders of magnitude faster than TLS, RSA, and PGP in encryption and decryption speed. Traditional encrypted messages require keys to be revoked manually, which can take minutes per user, while QSMS can revoke within 0.5 seconds. The traditional encryption method does not work for eavesdropping detection, while QSMS monitors QBER to detect and mitigate attacks. QBER-based intrusion detection is not available for traditional encryption method which are prone to be attacked by any cyber threats without being detected. QSMS thus effectively countermeasures the eavesdropping attack by decreasing the confidential quantum bit error rate (QBER) from 6.2% to 1.54%, providing a security recovery. These results show that QSMS provides a scalable and viable solution for email security in the post-quantum age. QSMS is a very resilient and future-proof cryptographic system since it allows one to identify cyber risks, cancel compromised keys, and stop key reuse.

V. CONCLUSION

In this work, we proposed the Quantum-Secured Mail System (QSMS), a hybrid encryption approach that combines application-layer encryption utilizing AES-256 with Quantum Key Distribution (QKD) to provide security for email communications against both classical and quantum adversaries. By employing the No-Cloning Theorem and Heisenberg Uncertainty Principle, QSMS guarantees a secure key exchange while preventing illegal copies and identifying live eavesdroppers. The experimental results demonstrated effective encryption with low processing cost and a high level of QBER-based intrusion detection. With a regime of key integrity enforcement and automated revocation, the scheme offered full protection against man-in-the-middle attacks on QKD and replay attacks on key exchange. With proven quantum security, QSMS is in a class above other encryption systems as a future-ready cryptographic tool.

Future work will focus on Quantum Key Distribution scalability and post-quantum cryptography (PQC) to increase the Quantum Security Management System (QSMS) for applications such real-time messaging and secure file transfer. Advanced quantum-resistant security models such as QSMS will become necessary for the protection of digital communications as quantum computing continues to evolve and proliferate.

REFERENCES

- [1] P. W. Shor, "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer," *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997.
- [2] D. J. Bernstein, T. Lange, and C. Peters, "Attacking and Defending the McEliece Cryptosystem," in *Post-Quantum Cryptography*, Springer, vol. 5299, 2008.
- [3] C. H. Bennett and G. Brassard, "Quantum Cryptography: Public Key Distribution and Coin Tossing," in *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, 1984.
- [4] A. Ekert, "Quantum Cryptography Based on Bell's Theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.
- [5] W. K. Wootters and W. H. Zurek, "A Single Quantum Cannot be Cloned," *Nature*, vol. 299, pp. 802–803, 1982.

- [6] N. Gisin et al., "Quantum Cryptography," *Reviews of Modern Physics*, vol. 74, no. 1, pp. 145–195, 2002.
- [7] C. H. Bennett and G. Brassard, "Quantum Cryptography: Public Key Distribution and Coin Tossing," *IEEE Int. Conf. Computers, Systems & Signal Processing*, 1984.
- [8] A. Ekert, "Quantum Cryptography Based on Bell's Theorem," *Phys. Rev. Lett.*, vol. 67, pp. 661–663, 1991.
- [9] M. Koashi, "Efficient Quantum Key Distribution with Practical Sources and Detectors," *Phys. Rev. A*, vol. 73, 2006.
- [10] L. O. Mailloux et al., "Performance Evaluations of QKD System Architecture," *IEEE Trans. Quantum Eng.*, vol. 1, 2015.
- [11] N. Gisin and G. Ribordy, "Quantum Computing and Cryptography," *Phys. Rev. Lett.*, vol. 74, 2007.
- [12] P. Arteaga-Díaz et al., "Side-Channel Attacks on Free-Space QKD Systems," *IEEE Access*, vol. 10, pp. 82697–82705, 2022.
- [13] W. Y. Hwang, "Quantum Key Distribution with High Loss," *Phys. Rev. A*, vol. 93, 2016.
- [14] W. K. Wootters and W. H. Zurek, "A Single Quantum Cannot be Cloned," *Nature*, vol. 299, 1982.
- [15] N. Gisin and S. Massar, "Optimal Quantum Cloning Machines," *Phys. Rev. Lett.*, vol. 79, 1997.
- [16] A. Boaron et al., "Secure Quantum Key Distribution Over 421 km," *Nat. Phys.*, vol. 14, 2018.
- [17] A. S. Cacciapuoti et al., "Quantum Internet: Networking Challenges," *IEEE Commun. Surveys Tuts.*, vol. 21, 2019.
- [18] Liao, S., Cai, W., Liu, W., Zhang, L., Li, Y., Ren, J., Yin, J., Shen, Q., Cao, Y., Li, Z., Li, F., Chen, X., Sun, L., Jia, J., Wu, J., Jiang, X., Wang, J., Huang, Y., Wang, Q., . . . Pan, J. (2017). Satellite-to-ground quantum key distribution. *Nature*, 549(7670), 43–47. <https://doi.org/10.1038/nature23655>
- [19] Cao, Y., Zhao, Y., Wang, Q., Zhang, J., Ng, S. X., & Hanzo, L. (2023). The evolution of quantum key distribution networks: On the road to the internet. *IEEE Communications Surveys & Tutorials*, 24(2), 839–894.
- [20] Arteaga-Díaz, P., Cano, D., & Fernandez, V. (2022). Practical side-channel attack on free-space QKD systems with misaligned sources and countermeasures. *IEEE Access*, 10, 82697–82705.

REFERENCES

- [1] M. Krenn, M. Huber, R. Fickler, R. Lapkiewicz, S. Ramelow, A. Zeilinger, Generation and confirmation of a (100×100) dimensional entangled quantum system. *Proc. Natl. Acad. Sci. U.S.A.* 111, 6243–6247 (2014).
- [2] A. C. Dada, J. Leach, G. S. Buller, M. J. Padgett, E. Andersson, Experimental high dimensional two-photon entanglement and violations of generalized Bell inequalities. *Nat. Phys.* 7, 677–680 (2011).
- [3] J. Harris, F. Bouchard, E. Santamato, W. H. Zurek, R. W. Boyd, E. Karimi, Quantum probabilities from quantum entanglement: Experimentally unpacking the Born rule. *New J. Phys.* 18, 053013 (2016).
- [4] F. Cardano, F. Massa, H. Qassim, E. Karimi, S. Slussarenko, D. Paparo, C. de Lisio, F. Sciarrino, E. Santamato, R. W. Boyd, L. Marrucci, Quantum walks and wavepacket dynamics on a lattice with twisted photons. *Sci. Adv.* 1, e1500087 (2015).
- [5] W. K. Wootters, W. H. Zurek, A single quantum cannot be cloned. *Nature* 299, 802–803 (1982). [6] N. Gisin, S. Massar, Optimal quantum cloning machines. *Phys. Rev. Lett.* 79, 2153 (1997).
- [7] D. Bruß, C. Macchiavello, Optimal state estimation for dimensional quantum systems. *Phys. Lett. A* 253, 249–251 (1999)
- [8] A. Lamas-Linares, C. Simon, J. C. Howell, D. Bouwmeester, Experimental quantum cloning of single photons. *Science* 296, 712–714 (2002 33)
- [9] W. T. M. Irvine, A. L. Linares, M. J. A. de Dood, D. Bouwmeester, Optimal quantum cloning on a beam splitter. *Phys. Rev. Lett.* 92, 047902 (2004).

- [10] P. Arteaga-Díaz, D. Cano and V. Fernandez, "Practical side channel attack on free-space QKD systems with misaligned sources and countermeasures", *IEEE Access*, vol. 10, pp. 82697-82705, 2022.
- [11] C.H. Bennett and Gilles Brassard, "Experimental quantum cryptography: the dawn of a new era for quantum cryptography: the experimental prototype is working", *ACM SIGACT News*, vol. 20, pp. 78-82, 1989.
- [12] S. Vittorio, "Quantum Cryptography: Privacy though Uncertainty", *CSA Discovery Guides*, 2002.
- [13] C. Matt and U. Maurer, "The one-time pad revisited", *IEEE Int. Symp. Inf. Theory - Proc.*, pp. 2706-2710, 2013.
- [14] G. Upadhyay and M. J. Nene, "One time pad generation using quantum superposition states", *2016 IEEE Int. Conf. Recent Trends Electron. Inf. Commun. Technol. RTEICT 2016 - Proc*, pp. 1882-1886, 2017.
- [15] Saeed, M. H., Sattar, H., Durad, M. H., & Haider, Z. (2022, August). Implementation of qkd bb84 protocol in qiskit. In *2022 19th International Bhurban Conference on Applied Sciences and Technology (IBCAST)* (pp. 689-695). IEEE.

